Online Appendix for Health and environmental impacts of ultra-processed food consumption in Argentina, a national modelling study

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Online Appendix

This Online Appendix accompanies the main manuscript and provides additional methodological details on data sources, processing steps, and the health, economic, and environmental modelling framework, together with supplementary tables.

Appendix A. Data sources and processing

A.1 Dietary intake data (ENNyS 2)

Dietary intake was obtained from Argentina's *Second National Survey of Nutrition and Health* (ENNyS 2, 2018–2019), a cross-sectional, multistage, region-stratified probabilistic survey conducted in urban localities. Dietary intake was collected with 24-hour recalls using a multiple-pass approach.[1,2]

In the survey file used for this study, 21,358 selected members were recorded overall. We restricted the analysis to adults aged 30–69 years, yielding 4,523 selected members in that age range. After linkage with the food-intake file, 4,496 adults aged 30–69 years had at least one recall with food-record information (1,888 men and 2,608 women) and therefore contributed to the dietary exposure analysis. Repeated recalls were available for 1,011 adults in this analytic age range. Recall-level intakes were first averaged within person across available recall days and then summarised with calibrated survey weights (w_k) to obtain population-representative estimates.

A.2 NOVA food classification

Each ENNyS 2 food and beverage code was classified according to the NOVA framework, which groups foods into four groups based on the extent and purpose of industrial processing rather than on nutrient content.[3] The four groups are: (1) unprocessed or minimally processed foods (NOVA 1); (2) processed culinary ingredients (NOVA 2); (3) processed foods (NOVA 3); and (4) ultra-processed foods (NOVA 4). Classification was performed at the individual food-code level using the Spanish item descriptions provided in the ENNyS 2 food file, supplemented where necessary by knowledge of typical Argentine product formulations. Dietary supplement codes (identified by codes beginning with the letter S) were excluded from the classification and from all analyses; the remaining 1,040 non-supplement codes are classified in Table S2 and their distribution across NOVA groups is summarised in Table S3.

The primary distinguishing criterion between groups is the presence or absence of industrial additives not normally used in home cooking and serving no culinary purpose beyond product palatability, preservation, or marketability. Concretely:

- **NOVA 1** encompasses foods that have undergone no industrial processing or only processes aimed at preservation without altering the essential nature of the product—such as fresh, chilled, or frozen single-ingredient vegetables, fruits, meats, fish, eggs, plain legumes, and plain whole grains. For example, fresh beef cuts, fresh chicken, whole-grain rice, raw eggs, and fresh seasonal vegetables were classified as NOVA 1.

- **NOVA 2** covers substances extracted from whole foods and used in home preparation—including

vegetable oils, sugar, salt, vinegar, and plain flour. These are not consumed on their own but are used to season or cook NOVA 1 foods.

- **NOVA 3** includes foods manufactured by applying salt, sugar, oil, vinegar, or fermentation to NOVA 1 foods to preserve them or modify their flavour, without the addition of industrial additives. Typical examples in the ENNyS 2 food list include canned or preserved vegetables (e.g., canned peas, bottled peppers in brine), cured or smoked cheeses and meats when formulated without flavour enhancers or stabilisers, canned fish in oil or brine, and commercially bottled fruit in syrup.

- **NOVA 4** (ultra-processed foods) encompasses industrial formulations that, in addition to NOVA 1 foods, sugars, oils, or salts, contain substances typically not found in home cooking—such as flavourings, emulsifiers, colour enhancers, stabilisers, artificial sweeteners, or high-fructose syrup—and/or food products assembled largely from industrial ingredients rather than whole foods. In the Argentine dietary context, this group included packaged biscuits, cookies, and crackers (*galletitas dulces, galletitas de agua, bizcochitos envasados*); packaged breads and pastries prepared from industrial mixes (*pan de molde, chipá de polvo premezcla*); breakfast cereals, cereal bars, and puffed-grain snacks; carbonated and sweetened non-alcoholic beverages; industrial fruit juices and flavoured drinks; spreadable processed cheeses; frankfurters, sausages, and formed meat products; instant noodles and dehydrated soups; confectionery including chocolates, sweets, and ice cream; flavoured yoghurts and dairy desserts; industrial sauces, dressings, and condiments with additives; and prepared frozen meals.

Several categories required judgement. Items sold both in artisan and industrial forms (e.g., *bizcochitos*, *empanadas*, *bizcochuelo*) were classified based on the ENNyS description: items explicitly described as packaged, industrial, or prepared from a premixed powder were assigned to NOVA 4, while those described as freshly prepared at home or in a bakery were assigned to NOVA 3 or examined case by case. Restaurant or food-service codes that could not be disaggregated into ingredients (e.g., fast-food items, complete restaurant dishes) were classified according to their predominant formulation and typical Argentine product characteristics. Where the description alone was insufficient, Argentine food labelling and product databases were consulted to infer the ingredient profile. Plain mate herb (*yerba mate*) was classified as NOVA 1; instant or flavoured mate preparations were classified as NOVA 4.
Of the 1,040 non-supplement codes, 310 (29.8%) were classified as NOVA 1, 69 (6.6%) as NOVA 2, 203 (19.5%) as NOVA 3, and 458 (44.0%) as NOVA 4 (Table S3). The high share of NOVA 4 codes reflects the diversity of packaged and industrialised products captured in ENNyS 2, though each code's contribution to total dietary energy depends on reported quantities consumed (Table S2).

A.3 Food composition and energy calculation

Food-specific energy density (kcal per 100 g) was obtained from the official composition resources linked to ENNyS 2. Item-level energy intake was computed as

$$E_{kji} = \frac{g_{kji}\,\mathrm{kcal}_i}{100},$$

where g_{kji} is grams of food code i consumed by individual k on recall day j, and kcal_i is kcal per 100 g.[1]

Food-code energy values were taken from the ENNyS 2-linked composition files, with an official secondary composition source used as a fallback when values were unavailable in the primary file. No additional item-level imputation rules were applied. When kcal_i was not available for a code, the corresponding item-level energy was treated as missing and excluded from item-level energy aggregation. Because the recall denominator uses the official ENNyS total energy variable, missing item-level energy affects only the reconstructed numerator used for E_{kj}^{UPF} and not the total-energy denominator itself. In practice, item-level energy coverage was high, so the potential impact of these missing values on derived %UPF estimates is expected to be small.[4]

A.4 Mortality data

Mortality inputs were obtained from Argentina's 2019 vital statistics (DEIS) and processed at record level using sex, age-group, underlying-cause, and death-count fields.[5] We retained records with known sex, excluded unspecified age-group records, and restricted the analysis to adults aged 30–69 years (groups 07_30 a 34 to 14_65 a 69), yielding 16 strata (two sexes by eight 5-year age groups). Death counts were aggregated within each stratum. The all-cause mortality input used in the CRA model was therefore based on all underlying-cause codes combined (no ICD-10 chapter restriction), with 103,448 deaths in the analytic population.

A.5 Population data

Population denominators were estimated from ENNyS 2 calibrated expansion factors and aggregated to the same 16 sex-by-age strata used in the mortality module (30–34 to 65–69 years, in men and women). To avoid duplicate contributions, each sampled person contributed once to the denominator, and only observations with valid positive total energy intake were retained. Stratum-specific population was computed as the sum of calibrated person-weights:

$$N_{sa} = \sum_{k \in (s,a)} w_k,$$

where w_k denotes the calibrated expansion factor for individual k. This procedure yields population-representative counts for the ENNyS 2 target frame (urban localities with $\geq$5,000 inhabitants).[1]
The resulting analytic population aged 30–69 years was 15,130,807, with non-zero population in each of the 16 sex-age strata.

A.6 Economic data

Economic inputs were obtained from Argentina's national household survey microdata for 2019 (fourth quarter). For each sex-age stratum, we estimated two labour-market parameters using survey weights: (i) mean labour income among employed individuals with positive income in their main occupation, and (ii) the employment rate (share employed within the stratum). Monthly labour income was annualised by multiplying by 12.[6]
These stratum-specific parameters were linked to attributable deaths and valued with a human-capital approach as the present value of lost earnings up to retirement age (65 years), applying a 3% annual discount rate. Final estimates were expressed in international dollars (intl.$) using the World Bank PPP conversion factor for Argentina in 2019 (20.38 ARS per intl.$).[7]

A.7 Environmental footprint data and food-code mapping

Environmental impacts were estimated by linking consumed food codes to life-cycle assessment coefficients for greenhouse gas emissions (GHG), land use, water use, and freshwater eutrophication. Coefficients were expressed per kilogram of product at a consistent system boundary; the hierarchy prioritised Argentina-focused evidence and was complemented with global datasets for groups not covered by primary sources.[8–11] Mapping coverage for in-scope foods is reported in Table S5, and the group-level coefficient set used in the manuscript environmental specification is reported in Table S8. Plain drinking water and ice were treated as out of scope for diet-related footprint accounting because they contribute negligible dietary energy and would distort mass-based coverage summaries.

Appendix B. Health, economic, and environmental modelling framework

Notation

Sex is indexed by s, age group by a (30–34, 35–39, ..., 65–69), individual by k, recall day by j, and food code by i. Calibrated survey weights are w_k. Observed deaths in stratum (s, a) are D_{sa} and UPF-attributable deaths are A_{sa}. The exposure is the dietary energy share from ultra-processed foods (UPFs), measured in percentage points from 0 to 100. Environmental indicators are indexed by m.

B.1 Estimation of UPF contribution to total energy intake

The exposure is defined as the percentage of total dietary energy from UPFs. First, UPF energy per recall is computed by summing item-level energy over food codes classified as NOVA 4 with official energy density available.

$$E_{kj}^{\mathrm{UPF}} = \sum_{i \in \mathcal{U}} \frac{g_{kji}\,\mathrm{kcal}_i}{100},$$

where $\mathcal{U}$ denotes the set of NOVA 4 food codes with available kcal_i.[1,3]
Second, recall-level %UPF uses the official total energy for that recall, E_{kj}^{tot}, as denominator.

$$\%\mathrm{UPF}_{kj} = 100 \times \frac{E_{kj}^{\mathrm{UPF}}}{E_{kj}^{\mathrm{tot}}}.$$

Recalls with non-positive total energy or values outside $[0, 100]$ are excluded.
Third, for participants with multiple recalls, we compute a within-person mean.

$$\overline{\%\mathrm{UPF}}_k = \frac{1}{J_k} \sum_{j=1}^{J_k} \%\mathrm{UPF}_{kj}.$$

Fourth, stratum-level summaries are computed as weighted moments.

$$\mu_{sa} = \frac{\sum_{k\in(s,a)} w_k \, \overline{\%\mathrm{UPF}}_k}{\sum_{k\in(s,a)} w_k}, \qquad \sigma_{sa} = \sqrt{\frac{\sum_{k\in(s,a)} w_k \left(\overline{\%\mathrm{UPF}}_k - \mu_{sa}\right)^2}{\sum_{k\in(s,a)} w_k}}.$$

Because full replicate-weight variance estimation was not available for all derived exposure moments, sampling uncertainty in μ_{sa} was propagated using an approximate standard error $\mathrm{SE}_{sa} = \sigma_{sa}/\sqrt{n_{sa}}$, where n_{sa} is the number of individuals in the stratum. This approximation should be interpreted as capturing within-stratum sampling variability rather than the full complex-survey variance.

B.2 Dose–response function and relative risk

The CRA framework requires a continuous risk function to translate the full stratum-specific exposure distribution (not only mean exposure) into expected risk. For this reason, we parameterised the UPF–mortality association as a log-linear dose–response function in which risk increases multiplicatively with each additional percentage point of dietary energy from UPFs:

$$\mathrm{RR}(p) = \exp(\beta p), \qquad \beta = \frac{\ln(\mathrm{RR}_{\mathrm{ref}})}{x_{\mathrm{ref}}}.$$

The reference effect size was taken from the meta-analysis used in the main model input, $\mathrm{RR}_{\mathrm{ref}} = 1.25$ (95% CI 1.14–1.37) for all-cause mortality.[12] To convert this pooled high-versus-low contrast into a continuous coefficient, we used the exposure contrast adopted in prior UPF CRA applications, $x_{\mathrm{ref}} = 35.7$ percentage points of energy from UPFs (interpreted as an approximately 0% to 35.7% contrast).[13] This calibration imposes two anchors: $\mathrm{RR}(0) = 1$ and $\mathrm{RR}(35.7) = 1.25$, yielding $\beta = \ln(1.25)/35.7 = 0.00625$.

Under this specification, relative risk at any exposure level can be evaluated directly and integrated over the observed exposure distribution. For interpretation, a 10-percentage-point increase in UPF energy share corresponds to $\exp(10\beta) = 1.064$, i.e., about a 6.4% higher mortality risk under the model assumptions.

Uncertainty in β was propagated from the reported confidence interval of $\mathrm{RR}_{\mathrm{ref}}$ by working on the log scale and assuming approximate normality of the log-risk coefficient, following standard burden-of-disease practice.[14] This uncertainty component was then sampled in the Monte Carlo stage (Section B.4).

B.3 Population attributable fraction (distributional CRA)

Population attributable fractions (PAFs) were computed using a distributional comparative risk assessment approach, integrating risk over the exposure distribution within each stratum.[13,14]

Within each stratum (s, a), exposure was modelled as log-normal, parameterised from the stratum mean and standard deviation.

$$X_{sa} \sim \mathrm{LogNormal}(\mu^*_{sa}, \sigma^{*2}_{sa}), \qquad \sigma^{*2}_{sa} = \ln\left(1 + \left(\frac{\sigma_{sa}}{\mu_{sa}}\right)^2\right), \quad \mu^*_{sa} = \ln(\mu_{sa}) - \frac{\sigma^{*2}_{sa}}{2}.$$

Because %UPF is bounded, expectations were evaluated over $[0, 100]$ using a truncated density.

$$\mathbb{E}[\mathrm{RR}(X_{sa})] = \int_0^{100} \exp(\beta x)\, f_{sa}(x)\, dx,$$

where f_{sa} denotes the truncated log-normal density.[14]
For a proportional reduction scenario with fraction r, exposure is shifted to $X_{sa}^{\mathrm{cf}} = (1-r)X_{sa}$ and

$$\mathrm{PAF}_{sa} = \frac{\mathbb{E}[\mathrm{RR}(X_{sa})] - \mathbb{E}[\mathrm{RR}(X_{sa}^{\mathrm{cf}})]}{\mathbb{E}[\mathrm{RR}(X_{sa})]}.$$

For a theoretical minimum risk level (TMRL) at 0% UPF, $\mathbb{E}[\mathrm{RR}(X_{sa}^{\mathrm{cf}})] = 1$.
Guideline-aligned scenarios imposed absolute %UPF targets derived from the National Dietary Guidelines (NDG) food-group structure.[15] For profile $q \in \{\text{conservative}, \text{central}, \text{strict}\}$, the target was constructed as

$$\%\mathrm{UPF}_q^{\mathrm{target}} = 100 \sum_g w_g\, c_{gq},$$

where w_g are normalised NDG basket composition weights ($\sum_g w_g = 1$) and c_{gq} are group-specific residual UPF caps under profile q. Profile multipliers were defined as an exploratory $\pm 50\%$ range around the central profile ($m_q \in \{1.5, 1.0, 0.5\}$), yielding targets of 5.45% (conservative), 3.63% (central), and 1.81% (strict). For each stratum, we set the counterfactual mean to $\mu_{saq}^{\mathrm{cf}} = \min(\mu_{sa}, \%\mathrm{UPF}_q^{\mathrm{target}})$ and scaled the standard deviation proportionally, $\sigma_{saq}^{\mathrm{cf}} = \sigma_{sa}\, \mu_{saq}^{\mathrm{cf}} / \max(\mu_{sa}, 0.001)$, before recomputing $\mathbb{E}[\mathrm{RR}(X_{sa}^{\mathrm{cf}})]$.
Attributable deaths were then computed as $A_{sa} = \mathrm{PAF}_{sa}\, D_{sa}$ and summed over strata to obtain national totals.[13,14]

Construction of group-specific UPF caps

The central cap c_g was set to reflect the maximum UPF fraction consistent with guideline-concordant consumption of that food group (Table B1). Food groups where whole or minimally processed forms are normative—fruits, vegetables, legumes, rice, starchy vegetables, eggs, and nuts—were assigned $c_g = 0$. Processed-grain groups (cereals, pasta), dairy products, and oil crops were assigned $c_g = 0.10$, allowing a modest share of processed variants (e.g., packaged pasta, flavoured yoghurt) while maintaining guideline adherence. Baked products received $c_g = 0.20$ because Argentina's guideline basket includes a small allowance for bread-type items, and NOVA 4 classification is common in this group. All animal-protein groups (meats, fish) were assigned $c_g = 0.02$, a near-zero allowance consistent with fresh-cut or lightly processed preparations. The three profiles apply a uniform multiplier: $c_{gq} = \min(c_g\, m_q,\, 0.95)$, where $m_{\mathrm{conservative}} = 1.5$, $m_{\mathrm{central}} = 1.0$, and $m_{\mathrm{strict}} = 0.5$.
Because the NDG normative diet assigns large energy shares to plant-based groups that carry a zero cap—fruits (18%), vegetables (25%), and legumes (5%)—the aggregated target is well below Argentina's observed UPF share of 16.66%, producing reductions of 67.3%, 78.2%, and 89.1% for the conservative, central, and strict profiles, respectively. The derivation of the 3.63% central target is shown in Table B1 and the scenario-level targets are summarised in Table S7.

Table B1. Derivation of the central NDG guideline-aligned UPF target (3.63% energy)

Food group	NDG weight w_g (%)	Central cap c_g	Contribution $w_g c_g$ (pp)
Fruits	18.0	0.00	0.00
Vegetables	25.0	0.00	0.00
Legumes and pulses	5.0	0.00	0.00
Cereals (excl. rice)	6.0	0.10	0.60
Rice	3.0	0.00	0.00
Pasta	3.0	0.10	0.30
Baked products	3.0	0.20	0.60
Starchy vegetables	3.0	0.00	0.00
Dairy products	13.0	0.10	1.30
Beef	4.0	0.02	0.08
Lamb and mutton	0.5	0.02	0.01
Other meats	0.5	0.02	0.01
Poultry	2.5	0.02	0.05
Pork	1.0	0.02	0.02
Fish and seafood	3.0	0.02	0.06
Eggs	1.5	0.00	0.00
Oil crops	6.0	0.10	0.60
Nuts and seeds	2.0	0.00	0.00
Total	**100.0**		**3.63**

NDG weights (w_g) are the normative energy shares prescribed by Argentina's 2016 National Dietary Guidelines. The central cap c_g is the maximum UPF fraction assigned to each food group for a guideline-concordant dietary pattern; caps are scenario assumptions intended to reflect guideline intent rather than observed values. The population-level target equals $100 \sum_g w_g c_g = 3.63\%$ of total dietary energy. Conservative (5.45%) and strict (1.81%) targets are obtained by multiplying all central caps by 1.5 and 0.5, respectively.

B.4 Monte Carlo uncertainty quantification

Because attributable mortality is a nonlinear function of exposure, risk coefficients, and death counts, analytic variance propagation is not straightforward in this setting. We therefore used probabilistic Monte Carlo simulation to jointly propagate uncertainty from key model inputs and obtain uncertainty intervals for both stratum-level and national estimates, following standard comparative risk assessment practice.[13,14]

We ran 10,000 iterations. For each stratum (s, a) and iteration b, inputs were sampled as:

$$\mu_{sa}^{(b)} \sim \mathcal{N}(\mu_{sa}, \mathrm{SE}_{sa}^2), \qquad \mu_{sa}^{(b)} \in [0.1, 100],$$

$$\beta^{(b)} \sim \mathcal{N}(\beta, \mathrm{SE}_{\beta}^2), \qquad D_{sa}^{(b)} \sim \mathrm{Poisson}(D_{sa}).$$

Here, $\mu_{sa}^{(b)}$ is the sampled mean exposure, truncated to the feasible %UPF range; $\beta^{(b)}$ is the sampled dose–response slope using the uncertainty derived from the confidence interval of the reference RR (Section B.2); and $D_{sa}^{(b)}$ is a sampled death count under a Poisson counting model. The stratum-specific exposure spread (σ_{sa}) was held at its estimated value and used to evaluate the distributional PAF under each draw.

For each iteration, we recomputed baseline PAFs, attributable deaths, and scenario-specific avertable deaths using the same equations as in Section B.3. Final point estimates are reported as medians (50th percentile),

and uncertainty intervals as the 2.5th and 97.5th percentiles of the simulated distributions.
To complement deaths averted, we report two additional mortality-burden metrics: years of life lost (YLL) and life expectancy gained. YLL captures premature mortality by weighting attributable deaths by remaining standard life expectancy, and life expectancy gain translates those avoided deaths into an intuitive population-level metric at age 30. Both measures follow standard WHO/GBD burden-of-disease conventions, improving comparability across studies and policy scenarios.[14,16]

B.5 Years of life lost (YLL)

YLL was estimated to capture the timing of premature mortality attributable to UPF intake, not only the number of attributable deaths. Death counts alone treat deaths at different ages as equivalent, while YLL assigns larger weight to deaths that occur earlier in adult life. This improves interpretability for prevention policy and keeps results comparable with WHO/GBD burden metrics.[14,16]
For each sex-age stratum, YLL is

$$\mathrm{YLL}_{sa} = A_{sa}\, L_a,$$

and national YLL is

$$\mathrm{YLL} = \sum_{s,a} A_{sa}\, L_a,$$

where A_{sa} are UPF-attributable deaths from the comparative risk assessment model (Sections B.2–B.4), and L_a denotes standard remaining life expectancy at age group a from the WHO reference table.[14,16]
In implementation, age-specific YLL weights were fixed for the 30–69 year groups (30–34: 60.34; 35–39: 55.38; 40–44: 50.43; 45–49: 45.51; 50–54: 40.61; 55–59: 35.74; 60–64: 30.92; 65–69: 26.21 years). These weights were applied identically across sexes, consistent with WHO/GBD practice of using a common standard life table for comparability.
No additional age weighting or time discounting was applied to YLL; burden differences are driven only by the number of attributable deaths and the standard life-years remaining in each age group.
Uncertainty was propagated at each Monte Carlo draw by recomputing

$$\mathrm{YLL}^{(b)} = \sum_{s,a} A_{sa}^{(b)} L_a.$$

For each scenario q, avertable YLL was computed as

$$\Delta\mathrm{YLL}^{(q,b)} = \sum_{s,a} A_{sa}^{(q,b)} L_a,$$

and final results were summarized as the median and 95% uncertainty interval (2.5th–97.5th percentiles) across draws (Section B.4).

B.6 Life expectancy gained (cause-deleted life table)

Life expectancy gain was included as a complementary mortality metric because it translates avertable deaths into expected years lived from age 30 under observed mortality conditions. In this framework, YLL quantifies

the burden removed, whereas the cause-deleted life table expresses the same removal as a shift in survival expectancy. This is a standard comparative risk assessment approach.[14,16–18]

A baseline life table was built from observed deaths and population in 5-year age bands (30–34 to 65–69). For each stratum, baseline central mortality was

$$m_{sa} = \frac{D_{sa}}{N_{sa}}.$$

Age-specific death probabilities were then obtained using the standard abridged conversion

$$q_x = \frac{n\, m_x}{1 + \frac{n}{2} m_x}, \qquad n = 5,$$

with a radix of $l_{30} = 100{,}000$. Survivors and deaths were updated recursively by $d_x = l_x q_x$ and $l_{x+n} = l_x - d_x$. Person-years in each interval were computed as

$$L_x = n\, l_{x+n} + \frac{n}{2} d_x,$$

then accumulated as $T_x = \sum_{y \geq x} L_y$, and life expectancy obtained as $e_x = T_x / l_x$.

For the final open interval, a terminal-life assumption was applied using expected years at age 70 of $E_{70} = 18.5$, so that the last age-group contribution includes survivors beyond 69 years.

The counterfactual removes UPF-attributable deaths,

$$D_{sa}^{\mathrm{cf}} = D_{sa} - A_{sa}, \qquad m_{sa}^{\mathrm{cf}} = \frac{D_{sa}^{\mathrm{cf}}}{N_{sa}},$$

and recomputes all life-table functions under the same assumptions. Life expectancy gain at age 30 is

$$\Delta e_{30} = e_{30}^{\mathrm{cf}} - e_{30}.$$

For scenario reporting, A_{sa} is replaced by scenario-specific avertable deaths $A_{sa}^{(q)}$, yielding $\Delta e_{30}^{(q)}$ for each policy target.

This cause-deleted construction assumes that removing UPF-attributable deaths changes all-cause mortality only through the deleted component, while other-cause hazards within each stratum remain unchanged. Under that assumption, the estimated Δe_{30} can be interpreted as a partial expectancy gain attributable to the modelled UPF mortality component in ages 30–69, rather than as a full forecast of national life expectancy dynamics.

Uncertainty in Δe_{30} was propagated by repeating the entire cause-deleted life table calculation at each Monte Carlo draw using $A_{sa}^{(b)}$ (or $A_{sa}^{(q,b)}$ for scenarios), and summarizing the resulting distribution with medians and 95% uncertainty intervals.

B.7 Indirect economic costs (human capital approach)

Indirect costs were included because mortality burden also implies loss of productive years, not only health loss. We used the human capital approach, in which each premature death is valued as the discounted stream of expected labour earnings that would have occurred until retirement age. This approach is widely

used in burden studies when the objective is to estimate the full production value forgone from premature mortality, and it is aligned with reference-case guidance when productivity effects are reported from a societal perspective.[19,20]

For each stratum (s, a), indirect costs were valued as the present value of forgone earnings until an assumed retirement age of 65 years for both sexes:

$$\mathrm{PVLE}_{sa} = A_{sa} \times y_{sa} \times emp_{sa} \times \sum_{t=1}^{T_{sa}} (1+r)^{-t},$$

where y_{sa} is mean annual labour income, emp_{sa} is the employment rate, $r = 0.03$ is the discount rate, and T_{sa} is remaining working years from the age-group midpoint (capped at zero). The annuity factor is

$$\sum_{t=1}^{T} (1+r)^{-t} = \frac{1-(1+r)^{-T}}{r}.$$

Age-midpoint mapping implies $T_{sa} = 0$ for 65–69 years, so deaths in that group contribute no additional labour-market losses under this framework.

In Argentina, statutory retirement ages are generally 60 years for women and 65 years for men; we used a uniform age-65 horizon as a simplifying assumption to apply a common valuation window across strata.[21] The 3% annual discount rate follows reference-case guidance.[19,20] Income and employment inputs were derived from national labour survey microdata (monthly income annualised by $\times 12$), and monetary totals were converted from ARS to intl.$ using the 2019 PPP factor (20.38 ARS/intl.$).[6,7,22]

The valuation is static and incidence-based: wages and employment rates are held at their observed 2019 stratum-specific values, with no additional assumptions on future wage growth, labour-force transitions, or macroeconomic feedback effects. Therefore, estimates should be interpreted as discounted market-income losses directly linked to modelled premature deaths under the observed labour market structure.

For each scenario q, avertable indirect cost was computed by replacing A_{sa} with $A_{sa}^{(q)}$ and summing over strata:

$$\Delta\mathrm{PVLE}^{(q)} = \sum_{s,a} A_{sa}^{(q)}\, y_{sa}\, emp_{sa}\, \frac{1-(1+r)^{-T_{sa}}}{r}.$$

Uncertainty was propagated by applying the same valuation equation to each simulation draw $A_{sa}^{(q,b)}$ and summarizing medians and 95% uncertainty intervals.

This specification captures market labour losses only. It does not include unpaid household production, caregiver time, employer replacement dynamics (friction-cost adjustments), or intangible welfare losses beyond labour income.

B.8 Environmental impacts, replacement matrix, and structural sensitivities

Environmental impacts were not assigned by building a separate life-cycle inventory for each ENNyS food code. Instead, each consumed ENNyS food code was linked through a mapping dictionary to an environmental food group, and group-level coefficients were then applied. Thus, multiple ENNyS codes can share the same environmental coefficient set. The environmental analysis used four indicators at the food-group level: greenhouse gas emissions (kg CO_2-eq), land use (m^2), water use (L), and freshwater eutrophication (g PO_4-eq)

per kg of reference mass.[8–11]

For each individual k, recall day j, and food code i, reported grams were first transformed into weighted population-representative quantities using the calibrated survey weight and dividing that weight by the number of recalls available for the same person. This yields weighted grams and weighted energy that can be aggregated without double counting repeated recalls. Positive-intake records were then linked to their NOVA group, environmental group, and energy density before impact coefficients were applied.

For each food code i and environmental indicator m, baseline item-level impact is

$$I_{kji,m} = q^{\text{ref}}_{kji}\, c_{i,m},$$

where q^{ref}_{kji} is the reference mass (kg) consistent with the coefficient definition and $c_{i,m}$ is the impact coefficient per kg. The term "reference mass" means the mass expressed in the same physical reference used by the environmental coefficient. This matters because intake is observed at the retail/consumption level in ENNyS, whereas some coefficients are defined at farm-gate level. When coefficients are defined at farm-gate level and observed quantities are recorded at retail level, mass is converted using a yield factor y_i (retail weight per farm weight).

$$q^{\text{ref}}_{kji} = \frac{q^{\text{ret}}_{kji}}{y_i}.$$

In practice, observed grams are first converted to retail kg and then, where necessary, to the mass reference required by the coefficient before multiplication by the corresponding per-kg impact factor. Baseline totals are then obtained by summing item-level impacts over all weighted records,

$$I^{\text{base}}_m = \sum_{k,j} \sum_{i} I_{kji,m},$$

and the UPF component is computed by restricting the same sum to NOVA 4 codes,

$$I^{\text{UPF}}_m = \sum_{k,j} \sum_{i \in \text{NOVA4}} I_{kji,m}.$$

The manuscript reports results from the environmental specification used for the main tables. In that specification, plain drinking water and ice were excluded from the environmental scope, because they dominate grams consumed but contribute no dietary energy, and selected beverages without a direct comparable-group match were reassigned to more appropriate beverage groups before coefficients were applied (e.g., mate/yerba, tea or infusions, coffee, industrial fruit juice, flavoured water, beer, wine, and spirits). Coverage diagnostics for this mapping are shown in Table S5. In this in-scope analysis, row-level mapping coverage was lower than gram- and energy-level coverage, but mapped energy coverage was high overall and within NOVA 4.

Counterfactual accounting. Counterfactual scenarios apply the same UPF-reduction fractions used in the health analysis (10%, 20%, 50%, and the NDG-aligned targets), denoted here by r. For each environmental indicator m, the accounting logic is the same in all scenarios: we first calculate the gross impact removed by reducing UPFs, then add back the impact of any replacement basket when substitution is assumed, and finally compare the resulting counterfactual total with the observed baseline.

Let the gross removed impact be

$$I_m^{\text{rem}} = r\, I_m^{\text{UPF}}.$$

Under no substitution, a fraction r of the UPF component is removed and nothing is added back. The counterfactual therefore becomes

$$I_m^{\text{cf}} = I_m^{\text{base}} - r\, I_m^{\text{UPF}}.$$

This scenario can be interpreted as a mechanical upper bound, because the removed UPF component is not replaced by any other food or beverage.

Under isocaloric substitution, the model removes a fraction r of mapped UPF intake but then restores the same amount of dietary energy using a non-UPF replacement basket. Let

$$E^{\text{red}} = r\, E_{\text{UPF}}^{\text{map}}$$

denote removed mapped UPF energy. For a replacement basket g, define the basket intensity per kcal as

$$\gamma_{m,g} = I_{m,g}/E_g,$$

that is, environmental impact per kcal of the replacement basket. The impact added back by replacement is

$$I_{m,g}^{\text{add}} = E^{\text{red}} \gamma_{m,g},$$

and the counterfactual total is

$$I_{m,g}^{\text{cf}} = I_m^{\text{base}} - r\, I_m^{\text{UPF}} + I_{m,g}^{\text{add}}.$$

In words, the net environmental effect under isocaloric substitution is the impact of the removed UPF energy minus the impact of the calories used to replace it. A basket with lower impact per kcal than the removed UPFs will generate a net reduction, whereas a basket with higher impact per kcal can attenuate or reverse that reduction.

As an alternative substitution structure, we also implemented isoweight substitution, replacing removed UPF mass one-for-one rather than replacing calories one-for-one. Let

$$M^{\text{red}} = r\, M_{\text{UPF}}^{\text{map}}$$

denote removed mapped UPF mass. For basket g, define impact intensity per kg as

$$\kappa_{m,g} = I_{m,g}/M_g.$$

The impact added back under mass-for-mass replacement is

$$I_{m,g}^{\text{add,kg}} = M^{\text{red}} \kappa_{m,g},$$

and the resulting counterfactual is

$$I_{m,g}^{\text{cf,W}} = I_m^{\text{base}} - r\, I_m^{\text{UPF}} + I_{m,g}^{\text{add,kg}}.$$

Because many non-UPF foods are less energy-dense than UPFs, isocaloric and isoweight substitution need not lead to the same environmental pattern.

We report environmental outcomes as net percentage change versus baseline,

$$\%\Delta I_m = 100\left(\frac{I_m^{\mathrm{cf}}}{I_m^{\mathrm{base}}} - 1\right),$$

so negative values indicate reductions and positive values indicate increases. Replacement baskets include NOVA 1, NOVA 3, a weighted NOVA 1/NOVA 3 mix, and an NDG-concordant basket. The NOVA 1 and NOVA 3 baskets use observed mapped non-UPF foods in those NOVA classes to estimate impact intensities. The weighted NOVA 1/NOVA 3 mix interpolates between those two structures using their observed mapped energy shares. The NDG-concordant basket uses food-group weights derived from the Argentine National Dietary Guidelines (NDG) together with observed mapped non-UPF foods to construct basket-specific impact intensities. The diet composition of these replacement baskets and the UPF targets used in the counterfactuals are summarised in Table S7. NOVA 2 was not used as a standalone replacement basket because its units and scope are not directly comparable to foods in diet-based footprint accounting.[15]

Appendix C. Supplementary tables

Tables S1–S8 summarise key inputs and supplementary results that are not duplicated in the main manuscript. Additional granular scenario outputs and mapping files are included in the project materials.

Table S1. Data sources summary

Input	Source	Year / resolution
Dietary intake (24h recalls)	ENNyS 2, Ministry of Health[1]	2018–2019 / individual
NOVA processing classification	NOVA framework[3]	food-code level
Relative risks (all-cause mortality)	Meta-analysis estimate with CRA calibration implementation[12,13]	published studies
Mortality counts by sex and age	Argentina vital statistics inputs; CRA framework[5,13,14]	2019 / strata
Economic inputs (income, employment)	National household survey microdata[6]	2019 / strata
PPP conversion factor	World Bank[7]	2019
Environmental coefficients	LCA sources and global datasets[8–11]	food-group level

Note. CRA denotes comparative risk assessment, PPP denotes purchasing power parity, and LCA denotes life-cycle assessment.

Table S2. NOVA classification of ENNyS food and beverage codes

Item description (Spanish)	Item description (English)	Code	NOVA
NOVA 1. Unprocessed or minimally processed foods			
Abadejo	Pout	P029	1
Acelga	Chard	V001	1
Acelga, congelada	Chard, frozen	V050	1
Acelga, pencas	Chard, stalks	V073	1
Acerola	Acerola cherry	F044	1
Achicoria	Chicory	V002	1
Achilate / achilata	Achilate / achilata drink	B055	1
Agua con gas / soda	Sparkling water/soda	B111	1
Agua de lluvia, río, canal, arroyo o acequia	Rainwater, river, canal, stream or ditch	B109	1
Agua de lluvia, río, canal, arroyo o acequia, hervida y/o con lavandina	Rainwater, river, canal, stream or ditch, boiled and/or with bleach	B110	1
Agua de mar	Seawater	B126	1
Agua de perforación con bomba	Borehole water (pumped)	B105	1
Agua de perforación con bomba, hervida y/o con lavandina	Borehole water (pumped), boiled and/or chlorinated (bleach)	B106	1
Agua de pozo	Well water	B107	1
Agua de pozo, hervida y/o con lavandina	Well water, boiled and/or with bleach	B108	1
Agua de red pública o agua corriente, dentro del hogar	Public mains tap water, in-home	B101	1
Agua de red pública o agua corriente, filtrada (filtro o dispenser de agua)	Public mains tap water, filtered (filter/dispenser)	B102	1
Agua embotellada (bidón / botella)	Bottled water (jerrycan/bottle)	B103	1
Agua embotellada (bidón / botella), bajo contenido de sodio	Bottled water (jerrycan/bottle), low sodium	B104	1
Agua origen desconocido	Water, source unknown	B100	1
Aguja	Chuck steak (aguja cut)	C078	1
Ají Locoto	Rocoto chili pepper (locoto)	V077	1
Ají rojo / morrón rojo	Red chili / red bell pepper	V003	1
Ají verde o amarillo / morrón verde o amarillo	Green or yellow chili pepper / green or yellow bell pepper	V004	1
Ajo	Garlic	V065	1
Albahaca	Basil	V071	1
Alcaucil	Artichoke	V005	1
Almeja	Clam	P042	1
Almendra	Almond	Z008	1
Amaranto	Amaranth	A002	1
Ananá	Pineapple	F003	1
Apio	Celery	V006	1
Arándanos	Blueberries	F060	1
Arroz blanco	White rice	A003	1
Arroz integral	Brown rice	A004	1
Arroz yamaní	Yamani rice	A102	1

Continued on next page

Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Arveja, congelada	Pea, frozen	V051	1
Arveja, fresca	Pea, fresh	V007	1
Arveja, semilla seca entera / partida	Pea, dried whole/split seed	A006	1
Asado con hueso***	Roast with bone***	C076	1
Asado sin hueso***	Boneless roast***	C075	1
Atún, lomito fresco	Tuna, fresh loin	P028	1
Avellana	Hazelnut	Z009	1
Avena, arrollada	Oats, rolled	A005	1
Banana	Banana	F006	1
Banana deshidratada	Dehydrated banana	F067	1
Batata	Sweet potato	A073	1
Berenjena	Eggplant	V010	1
Berro	Watercress	V011	1
Bife angosto con hueso***	Striploin steak, bone-in***	C089	1
Bife angosto sin hueso***	Striploin steak, boneless***	C088	1
Bola de lomo	Eye of round	C082	1
Brócoli	Broccoli	V012	1
Brotes de soja	Soybean sprouts	V013	1
Brótola	Brotola	P021	1
Caballa, tronco	Mackerel, trunk	P011	1
Cabra	Goat	C001	1
Café en saquitos (ingresar cantidad de saquitos)*	Coffee in bags (enter number of bags)*	B073	1
Café preparado a partir de grano molido (tipo café de filtro)	Coffee prepared from ground beans (filter coffee type)	B058	1
Calamar	Squid	P003	1
Caldo de verduras colado	Strained vegetable broth	V082	1
Camarón	Shrimp	P004	1
Caracú	Beef marrow bone (caracú)	C109	1
Carne picada común	Regular ground beef	C073	1
Carne picada especial	Lean ground beef	C074	1
Carpincho	Capybara	C050	1
Cayota o Alcayota	Cayota or Alcayota	F045	1
Cebolla	Onion	V014	1
Cebolla de verdeo	Green onion	V052	1
Cerdo bondiola	Pork neck (bondiola)	C100	1
Cerdo costillar tipo "ribs"	Pork spare ribs	C099	1
Cerdo costillita	Pork short ribs	C098	1
Cerdo jamón (corte de carne)	Pork leg/fresh ham (cut of meat)	C104	1
Cerdo matambre	Pork flank (matambre)	C101	1
Cerdo paleta (corte de carne)	Pork shoulder (cut of meat)	C105	1
Cerdo pechito	Pork belly	C102	1
Cerdo, patas, patitas	Pork trotters	C115	1
Cerdo, PROMEDIO	Pork, AVERAGE	C002	1
Cereza	Cherry	F007	1
Champignones, frescos	Mushrooms, fresh	V076	1
Charata (ave)	Charata (bird)	C064	1

Continued on next page

Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Chaucha, congelada	Green beans, frozen	V053	1
Chaucha, fresca	Green beans, fresh	V016	1
Chinchulines, tripa gorda	Chitterlings (small intestines), large intestine	C007	1
Chivito	Goat kid meat	C110	1
Chivito, cabeza	Chivito, head	C129	1
Choclo	Corn	A076	1
Choclo, congelado en grano	Corn, frozen grain	A176	1
Chuchu o chayote	Chuchu or chayote	V078	1
Ciruela	Plum	F008	1
Ciruela pasa / ciruela seca	Prune / dried plum	F009	1
Coco fresco	Fresh coconut	F046	1
Coco, leche	Coconut, milk	F066	1
Coliflor	Cauliflower	V015	1
Colita de cuadril	Rump tail (colita de cuadril)	C093	1
Conejo	Rabbit	C008	1
Corazón, vacuno	Heart, beef	C107	1
Cordero, PROMEDIO	Lamb, AVERAGE	C009	1
Cornalitos	Cornalitos (small silverside fish)	P031	1
Corvina	Croaker	P023	1
Cous Cous	Couscous	A118	1
Cuadrada	Cuadrada (beef bottom round)	C084	1
Cuadril	Rump (cuadril)	C092	1
Cuerito de cerdo	Pig skin	C052	1
Damasco	Apricot	F010	1
Dátiles	Dates	F061	1
Dorado	Golden dorado (river fish)	P035	1
Durazno	Peach	F011	1
Durazno orejón	Dried peach	F014	1
Escarola	Endive	V020	1
Espárrago	Asparagus	V021	1
Espinaca	Spinach	V022	1
Espinazo, vacuno	Backbone, beef	C120	1
Falda	Skirt/plate steak (falda)	C079	1
Frambuesa	Raspberry	F064	1
Frutilla	Strawberry	F015	1
Garbanzos	Chickpeas	A025	1
Garza	Heron	C062	1
Gatuzo	Smoothhound shark (gatuzo)	P024	1
Gérmen de trigo	Wheat germ	A129	1
Granada	Pomegranate	F016	1
Guabirova	Guabiroba (native berry)	F053	1
Guanaco, carne	Guanaco, meat	C111	1
Guayaba	Guava	F017	1
Habas	Broad beans	V023	1
Hakusai o col china	Hakusai or Chinese cabbage	V054	1
Hielo para bebidas	Ice for drinks	B114	1
Hígado	Liver	C012	1

Continued on next page

Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Hígado de pollo	Chicken liver	C122	1
Higo	Fig	F018	1
Higo desecado	Dried fig	F054	1
Hinojo	Fennel	V024	1
Hojas de parra	Grape leaves	V083	1
Hongos, frescos	Mushrooms, fresh	V025	1
Hongos, secos	Mushrooms, dried	V080	1
Huevo de codorníz, entero	Quail egg, whole	H004	1
Huevo de gallina, clara	Chicken egg, white	H002	1
Huevo de gallina, entero	Chicken egg, whole	H001	1
Huevo de gallina, yema	Chicken egg, yolk	H003	1
Infusión de hierbas en hebras tipo Boldo, Manzanilla, otros (ingresar cantidad en	Infusion of herbs in strands such as Boldo, Chamomile, others (enter quantity in	B127	1
Infusión de hierbas en saquitos tipo Boldo, Manzanilla, otros (ingresar cantidad de	Herbal infusion in bags such as Boldo, Chamomile, others (enter quantity of	B068	1
Jabalí	Wild boar	C103	1
Jacaré	Jacare	C061	1
Jengibre	Ginger	V066	1
Jugo de arándanos, natural	Cranberry juice, natural	B124	1
Jugo de fruta, exprimido sin azúcar agregada, envasado	Fruit juice, squeezed without added sugar, packaged	B032	1
Jugo de fruta, pasteurizado no reconstituido	Fruit juice, pasteurized not reconstituted	B033	1
Jugo de limón, exprimido	Lemon juice, squeezed	B034	1
Jugo de naranja, exprimido	Orange juice, squeezed	B035	1
Jugo de papaya, natural	Papaya juice, natural	B116	1
Jugo de pomelo, exprimido	Grapefruit juice, squeezed	B036	1
Kaki	Persimmon	F019	1
Kale	Kale	V057	1
Kinoto	Kumquat	F020	1
Kiwi	Kiwi	F021	1
Lampalagua (serpiente)	Lampalagua (snake)	C065	1
Langostino	Prawn	P005	1
Leche de cabra, entera, en polvo*	Goat milk, whole, powdered*	L070	1
Leche de cabra, entera, fluida	Goat milk, whole, fluid	L002	1
Leche de oveja, entera, fluida	Sheep's milk, whole, fluid	L003	1
Leche de vaca, ordeñada (sin ningún tipo de procesamiento)	Cow's milk, milked (without any processing)	L067	1
Leche descremada en polvo, fortificada con vitaminas A y D*	Skim milk powder, fortified with vitamins A and D*	L008	1
Leche descremada fluida, con 50% más de proteínas	More protein	L060	1
Leche en polvo, 25% menos de grasas saturadas, fortificada con vitaminas A, D, C, calcio y zinc*	Less saturated fat, fortified with vitamins A,	L045	1

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Leche entera en polvo, deslactosada, fortificada con vitaminas A y D*	Whole milk powder, lactose-free, fortified with vitamins A and D*	L009	1
Leche entera en polvo, fortificada con hierro*	Whole milk powder, fortified with iron*	L019	1
Leche entera en polvo, fortificada con vitaminas A y D*	Whole milk powder, fortified with vitamins A and D*	L047	1
Leche entera en polvo, fortificada con vitaminas A y D, para desayunos	Whole milk powder, fortified with vitamins A and D, for breakfast	L046	1
Leche entera en polvo, Plan Materno Infantil / Programas Sociales / otros, fortificada con hierro, zinc y ác. ascórbico*	Fortified with iron, zinc and acid. ascorbic*	L033	1
Leche entera en polvo, Plan Materno Infantil, sin fortificación*	Whole milk powder, Maternal and Child Plan, without fortification*	L032	1
Leche entera en polvo, Plan Vida, Ministerio de Desarrollo Social Pcia. de Bs. As., fortificada con vitaminas A y D*	As., fortified with vitamins A and D*	L043	1
Leche entera en polvo, sin fortificación*	Whole milk powder, unfortified*	L010	1
Leche entera fluida, con vitaminas A y D	Fluid whole milk, with vitamins A and D	L007	1
Leche entera fluida, deslactosada, fortificada con vitaminas A y D	Fluid, lactose-free whole milk, fortified with vitamins A and D	L011	1
Leche entera fluida, fortificada con hierro	Fluid whole milk, fortified with iron	L020	1
Leche entera fluida, fortificada con vitaminas A, D y B9	Fluid whole milk, fortified with vitamins A, D and B9	L048	1
Leche entera fluida, fortificada con vitaminas A, D, C, B9 y extra calcio	Fluid whole milk, fortified with vitamins A, D, C, B9 and extra calcium	L054	1
Leche entera fluida, sin fortificación	Fluid whole milk, without fortification	L006	1
Leche materna	Breast milk	L040	1
Leche parcialmente descremada fluida, deslactosada, fortificada con vitaminas	Fluid, lactose-free, partially skimmed milk, fortified with vitamins	L012	1
Leche parcialmente descremada fluida, fortificada con hierro	Partially skimmed milk, fortified with iron	L031	1
Leche parcialmente descremada fluida, fortificada con vitaminas A y D	Fluid partially skimmed milk, fortified with vitamins A and D	L005	1
Leche parcialmente descremada fluida, fortificada con vitaminas A, D, C y ác.	Fluid partially skimmed milk, fortified with vitamins A, D, C and AC.	L030	1

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Leche parcialmente descremada fluida, sin fortificación	Fluid partially skimmed milk, without fortification	L004	1
Leche semidescremada en polvo, deslactosada*	Semi-skimmed milk powder, lactose-free*	L041	1
Lechuga	Lettuce	V027	1
Lengua	Beef tongue	C013	1
Lenguado	Sole	P022	1
Lentejas	Lentils	A035	1
Liebre	Hare	C127	1
Limón	Lemon	F022	1
Limón, ralladura	Lemon, zest	F059	1
Llama	Llama meat	C128	1
Lomo	Beef tenderloin (lomo)	C085	1
Maíz, grano entero	Corn, whole grain	A036	1
Mamón	Papaya (mamon)	F023	1
Mandarina	Tangerine	F024	1
Mandioca	Cassava	V028	1
Mango	Mango	F025	1
Maní tostado sin sal	Unsalted roasted peanuts	Z010	1
Manzana con piel	Apple with skin	F027	1
Manzana sin piel	Apple without skin	F028	1
Maracuyá	Passion fruit	F070	1
Matambre (corte de carne)	Matambre (thin flank steak)	C080	1
Mate cocido en saquitos (ingresar cantidad de saquitos)*	Mate cocido tea bags (enter number of tea bags)*	B065	1
Mc Donald's, Cajita Feliz, Tomate cherry 50g (ingresar cantidad de porciones)	S, Happy Meal, Cherry Tomato 50g (enter number of servings)	R044	1
Mejillón	Mussel	P006	1
Melón	Melon	F029	1
Membrillo	Quince	F030	1
Merluza	Hake	P020	1
Mix de frutos secos, semillas y pasas de uva	Mix of nuts, seeds and raisins	Z016	1
Mollejas	Sweetbread	C014	1
Mondongo	Tripe	C015	1
Mora	Blackberry	F055	1
Nalga	Beef topside (nalga)	C081	1
Naranja	Orange	F031	1
Naranja, ralladura	Orange, zest	F062	1
Níspero	Medlar	F032	1
Nuez	Nut	Z012	1
Nutria	Otter	C112	1
Osobuco	Osso buco (cross-cut shank)	C108	1
Pacú	Pacu	P036	1
Paleta (corte de carne)	Shoulder (cut of meat)	C087	1
Palometa	Pomfret (palometa)	P033	1
Palomita (corte de carne)	Palomita (cut of meat)	C091	1

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Palta	Avocado	F034	1
Papa	Potato	A079	1
Papaya	Papaya	F063	1
Pato	Duck	C019	1
Pavo	Turkey	C020	1
Peceto	Eye of round roast (peceto)	C083	1
Pejerrey	Silverside	P030	1
Pepino	Cucumber	V031	1
Pera	Pear	F036	1
Pera orejón	Dried pear	F039	1
Perdiz	Partridge	C063	1
Perejil	Parsley	V067	1
Pescados de mar (ej: brótola, corvina blanca, gatuso, lenguado, merluza, etc.),	Sea fish (e.g. brotola, white croaker, catfish, sole, hake, etc.),	P001	1
Pescados de río (ej: dorado, palometa, surubí, trucha, etc.), PROMEDIO	River fish (e.g. dorado, palometa, surubí, trout, etc.), AVERAGE	P002	1
Pistacho	Pistachio	Z013	1
Pitanga	Surinam cherry (pitanga)	F056	1
Pollo con piel	Chicken with skin	C021	1
Pollo de mar	Sea chicken	P025	1
Pollo sin piel	Skinless chicken	C023	1
Pollo, alitas	Chicken, wings	C125	1
Pollo, menudos	Chicken, giblets	C022	1
Pollo, pechuga sin piel	Chicken, skinless breast	C059	1
Pomelo	Grapefruit	F041	1
Porotos	Beans	A051	1
Puerro	Leek	V032	1
Pulpo	Octopus	P038	1
Quinoa, semilla	Quinoa, seed	A053	1
Rabanito	Radish	V033	1
Radicheta	Wild chicory (radicheta)	V034	1
Rana	Frog	C117	1
Remolacha	Beet	V035	1
Repollito de Bruselas	Brussels sprouts	V036	1
Repollo	Cabbage	V037	1
Riñón	Kidney	C025	1
Roast beef	Roast beef cut	C086	1
Rúcula	Arugula	V058	1
Sábalo	Shad	P037	1
Salmón blanco	White salmon	P026	1
Salmón rosado	Pink salmon	P027	1
Salvado de avena	Oat bran	A055	1
Salvado de trigo	Wheat bran	A056	1
Sandía	Watermelon	F042	1
Semilla de amapola	Poppy seed	Z046	1
Semilla de chía	Chia seed	Z041	1

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Semilla de girasol	Sunflower seed	Z014	1
Semilla de lino	Flax seed	Z039	1
Semilla de sésamo	Sesame seed	Z015	1
Semilla de zapallo	Pumpkin seed	Z044	1
Sesos	Brains	C029	1
Soja, porotos	Soybeans	A052	1
Surubí	Surubi catfish	P034	1
Tatú	Armadillo (tatú)	C096	1
Té blanco en hebras (ingresar cantidad en gramos)*	White tea in threads (enter quantity in grams)*	B076	1
Té blanco en saquitos (ingresar cantidad de saquitos)*	White tea in bags (enter number of bags)*	B075	1
Té negro en hebras (ingresar cantidad en gramos)*	Black tea in threads (enter quantity in grams)*	B071	1
Té negro en saquitos (ingresar cantidad de saquitos)*	Black tea in bags (enter number of bags)*	B066	1
Té verde en saquitos (ingresar cantidad de saquitos)*	Green tea in bags (enter number of bags)*	B067	1
Tofu	Tofu	A168	1
Tomate	Tomato	V038	1
Tomate cherry	Cherry tomato	V059	1
Tortuguita	Tortuguita (beef cut)	C090	1
Trigo burgol	Bulgur wheat	A239	1
Trigo, grano entero	Wheat, whole grain	A060	1
Trucha	Trout	P032	1
Tuna	Prickly pear	F058	1
Ubre	Udder	C030	1
Uva	Grape	F043	1
Uva pasa	Raisin	F035	1
Vacío	Beef flank (vacío)	C077	1
Vacuno, cortes grasos, con hueso (ej: asado), PROMEDIO	Beef, fatty cuts, with bone (e.g. roast), AVERAGE	C032	1
Vacuno, cortes grasos, sin hueso (ej: vacío, carne picada común, aguja, falda, alita,	Beef, fatty cuts, boneless (e.g. vacío, regular ground beef, chuck, skirt, wing,	C031	1
Vacuno, cortes magros, con hueso, PROMEDIO	Beef, lean cuts, bone-in, AVERAGE	C034	1
Vacuno, cortes magros, sin hueso (ej: nalga, bola de lomo, peceto, cuadrada, picada intensamente desgrasada), PROMEDIO	Intensely defatted), AVERAGE	C033	1
Vacuno, cortes semigrasos, con hueso (ej: bife angosto), PROMEDIO	Beef, medium-fat cuts, bone-in (e.g., striploin), average	C036	1
Vacuno, cortes semigrasos, sin hueso (ej: lomo, carne picada especial, roast beef, paleta, bife angosto, tortuguita, palomita), PROMEDIO	Shoulder, striploin, tortuguita and palomita cuts), average	C035	1

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Verdolaga	Purslane	V079	1
Vizcacha	Viscacha (wild rodent meat)	C060	1
Yerba mate para mate cebado (ingresar cantidad en gramos)*	Yerba mate for brewed mate (enter grams)*	B064	1
Yerba mate para mate cocido (ingresar cantidad en gramos)*	Yerba mate for mate cocido infusion (enter grams)*	B070	1
Yerba mate para tereré (ingresar cantidad en gramos)*	Yerba mate for terere infusion (enter grams)*	B074	1
Yogur entero natural	Natural whole yogurt	Y001	1
Zanahoria	Carrot	V040	1
Zapallito	Zucchini	V041	1
Zapallo	Pumpkin	V042	1
Zucchini	Zucchini	V061	1
NOVA 2. Processed culinary ingredients			
Aceite comestible mezcla	Edible oil mixture	Z003	2
Aceite comestible mezcla, para cocción	Edible oil mixture, for cooking	Z030	2
Aceite de canola	Canola oil	Z005	2
Aceite de canola, para cocción	Canola oil, for cooking	Z032	2
Aceite de chía	Chia oil	Z023	2
Aceite de chía, para cocción	Chia oil, for cooking	Z036	2
Aceite de coco	Coconut oil	Z022	2
Aceite de coco, para cocción	Coconut oil, for cooking	Z034	2
Aceite de girasol	Sunflower oil	Z001	2
Aceite de girasol alto oleico	High oleic sunflower oil	Z020	2
Aceite de girasol alto oleico, para cocción	High oleic sunflower oil, for cooking	Z027	2
Aceite de girasol, para cocción	Sunflower oil, for cooking	Z026	2
Aceite de maíz	Corn oil	Z002	2
Aceite de maíz, para cocción	Corn oil, for cooking	Z028	2
Aceite de oliva	Olive oil	Z004	2
Aceite de oliva, para cocción	Olive oil, for cooking	Z031	2
Aceite de sésamo	Sesame oil	Z042	2
Aceite de soja	Soybean oil	Z007	2
Aceite de soja, para cocción	Soybean oil, for cooking	Z035	2
Aceite de uva	Grapeseed oil	Z006	2
Aceite de uva, para cocción	Grapeseed oil, for cooking	Z033	2
Aceite mezcla con oliva	Olive oil blend	Z021	2
Aceite mezcla con oliva, para cocción	Olive oil blend, for cooking	Z029	2
Aceite para cocción, PROMEDIO	Cooking oil, AVERAGE	Z025	2
Aceite, PROMEDIO	Oil, AVERAGE	Z000	2
Almidón de maíz	Corn starch	A001	2
Almidón de mandioca	Cassava starch	A193	2
Arrope	Arrope (concentrated fruit syrup)	D026	2
Azúcar blanca molida	Ground white sugar	D003	2
Azúcar impalpable	Powdered sugar	D127	2

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Azúcar morena / azúcar mascabo	Brown sugar / mascabo sugar	D004	2
Bicarbonato de sodio	Baking soda	O042	2
Cacao amargo*	Bitter cocoa*	D050	2
Cacao en polvo sin fortificar*	Unfortified cocoa powder*	D018	2
Crema de leche	Heavy cream	G013	2
Glucosa	Glucose	D116	2
Gomasio (sésamo integral tostado y molido con sal marina)	Gomasio (toasted and ground whole sesame with sea salt)	Z045	2
Grasa de cerdo	Pork fat	G008	2
Grasa de pollo	Chicken fat	G019	2
Grasa vacuna	Beef fat	G009	2
Harina de algarrobo	Carob flour	A027	2
Harina de arroz	Rice flour	A028	2
Harina de cebada	Barley flour	A029	2
Harina de centeno	Rye flour	A030	2
Harina de chía	Chia flour	A212	2
Harina de garbanzos	Chickpea flour	A131	2
Harina de maíz	Cornmeal	A031	2
Harina de quinoa	Quinoa flour	A070	2
Harina de soja	Soy flour	A032	2
Harina de trigo	Wheat flour	A033	2
Harina de trigo fortificada	Fortified wheat flour	A132	2
Harina de trigo integral	Whole wheat flour	A034	2
Harina leudante	Self-rising flour	A133	2
Kero o melaza	Kero or molasses	D027	2
Levadura en polvo	Baking powder	A178	2
Levadura fresca	Fresh yeast	A179	2
Manteca	Butter	G010	2
Miel	Honey	D011	2
Miel de caña	Cane honey	D134	2
Polvo de hornear	Baking powder	A190	2
Rapadura	Rapadura (unrefined cane sugar)	D105	2
Sal	Salt	O026	2
Sal Marina	Sea Salt	O050	2
Sal modificada, 33% Cloruro de sodio	Sodium chloride	O031	2
Sal modificada, 66% Cloruro de sodio	Sodium chloride	O032	2
Sémola	Semolina	A058	2
Sémola fortificada	Fortified semolina	A195	2
Tapioca (almidón de mandioca)	Tapioca (cassava starch)	A059	2
Vinagre	Vinegar	O023	2
NOVA 3. Processed foods			
Aceituna negra	Black olive	F050	3
Aceituna verde	Green olive	F001	3
Ají en conserva	Canned chili	V081	3
Alcaparras, encurtido	Capers, pickled	V069	3
Almíbar	Syrup	D100	3

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Ananá, enlatado (fruta y almíbar)	Pineapple, canned (fruit and syrup)	F004	3
Anchi	Cornmeal porridge (anchi)	A101	3
Anchoa, enlatada en aceite	Anchovy, canned in oil	P010	3
Aperitivo, tipo Gancia	Appetizer, Gancia type	B119	3
Arveja, enlatada	Pea, canned	V008	3
Atún, enlatado al natural	Tuna, canned in water	P007	3
Atún, enlatado en aceite	Tuna, canned in oil	P008	3
Bizcochos de grasa (panadería) / Criollitos, Libritos (pan con grasa)	Fat biscuits (bakery) / Criollitos, Libritos (bread with fat)	A009	3
Bizcochos tipo Bay Biscuit	Bay Biscuit	A010	3
Bollitos de anís	Anise buns	A113	3
Brotes de tacuara (bambú) en escabeche	Pickled tacuara (bamboo) shoots	V075	3
Brownie (panadería)	Brownie (bakery)	A236	3
Butifarra	Butifarra (white sausage)	C123	3
Caballa, enlatada	Mackerel, canned	P012	3
Capelettis frescos, artesanal	Fresh, artisanal capelettis	A011	3
Cerveza con alcohol	Alcoholic beer	B006	3
Cerveza sin alcohol	Non-alcoholic beer	B007	3
Champagne	Sparkling wine (champagne-style)	B042	3
Champignones, enlatados	Mushrooms, canned	V074	3
Charqui de cordero o chalona	Lamb jerky or chalone	C038	3
Charqui de vaca	Cow jerky	C037	3
Cheesecake (panadería / confitería)	Cheesecake (bakery/confectionery)	A248	3
Chicha, bebida tradicional obtenida por fermentación del maíz	Chicha, traditional drink obtained by fermenting corn	B128	3
Chicharrón	Crackling	G017	3
Chipá	Chipa cheese bread	A013	3
Chipaco o pan con chicharrón, comprado	Chipaco or bread with chicharrón, purchased	A216	3
Choclo, enlatado cremoso	Corn, creamy canned	A078	3
Choclo, enlatado en grano	Corn, canned in grain	A077	3
Chorizo	Chorizo (fresh pork sausage)	C003	3
Chorizo seco	Dry chorizo	C095	3
Churros	Churros	A014	3
Coco rallado	Grated coconut	F051	3
Coctail de frutas, enlatado (fruta y almíbar)	Fruit cocktail, canned (fruit and syrup)	F052	3
Coquitos (panadería)	Coquitos (bakery)	A204	3
Cucurucho / cubanito	Wafer cone / cubanito wafer roll	A119	3
Dulce de batata	Sweet potato paste	D005	3
Dulce de cayote	Cayote jam	D073	3
Dulce de mamón	Papaya jam (mamón)	D120	3
Dulce de membrillo	Quince paste	D006	3

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Dulce de zapallo (compacto)	Pumpkin preserve (compact)	D128	3
Durazno, enlatado (fruta y almíbar)	Peach, canned (fruit and syrup)	F012	3
Empanaditas chinas (restaurant)	Chinese dumplings (restaurant)	A250	3
Facturas rellenas	Filled pastries (facturas)	A015	3
Facturas simples	Plain pastries (facturas)	A016	3
Fideos de arroz, SIN TACC	Rice noodles, TACC-FREE	A207	3
Fideos frescos, al huevo	Fresh noodles, with egg	A017	3
Fideos secos	Dry noodles	A018	3
Fideos secos integrales / fideos secos de sémola y harina de legumbres	Whole grain dry noodles / semolina and legume flour dry noodles	A122	3
Fideos secos, fortificados	Dry noodles, fortified	A121	3
Frangollo (maíz seco pisado)	Frangollo (dried crushed corn)	A201	3
Garbanzos, enlatados	Chickpeas, canned	A128	3
Grisines	Breadsticks (grissini)	A069	3
Grisines sin sal	Grisines without salt	A130	3
Hamburguesa de pescado (pescadería)	Fish burger (fishmonger)	P046	3
Hongos, enlatados	Mushrooms, canned	V026	3
Jalea tipo membrillo	Quince type jelly	D007	3
Jamón crudo	Dry-cured ham (jamón crudo)	C005	3
Jardinera, congelada	Mixed vegetables, frozen	V055	3
Jardinera, enlatada	Mixed vegetables, canned	V056	3
Leberwurst	Liverwurst (leberwurst)	C054	3
Leche condensada	Condensed milk	L034	3
Leche entera fluida, con azúcar, lista para consumir	Fluid whole milk, with sugar, ready to consume	L039	3
Lentejas, enlatadas	Lentils, canned	A134	3
Lomito ahumado	Smoked tenderloin	C056	3
Longaniza	Longaniza (fresh pork sausage)	C055	3
Maní tostado salado	Salted roasted peanuts	Z011	3
Masa de tarta o empanada, integral	Pie or empanada dough, whole wheat	A252	3
Masa de tarta o empanadas	Pie or empanada dough	A037	3
Masa de tarta, SIN TACC	Pie dough, WITHOUT TACC	A218	3
Masas finas surtidas (panadería / confitería)	Assorted thin doughs (bakery / confectionery)	A242	3
Masas secas surtidas (panadería / confitería)	Assorted dry doughs (bakery / confectionery)	A232	3
Merluza, enlatada en agua y aceite	Hake, canned in water and oil	P045	3
Milanesa de carne vacuna rebozada (carnicería)	Breaded beef Milanese (butcher shop)	C106	3
Milanesa de cerdo (carnicería)	Pork Milanese (butcher shop)	C126	3
Milanesa de pollo rebozada (pollería / granja)	Breaded chicken Milanese (poultry / farm)	C068	3
Morcilla	Black pudding	C016	3

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Muzzarella rebozada sin freir (pollería o granja)	Battered mozzarella without frying (poultry or farm)	Q048	3
Ñoquis de papa frescos, artesanal	Fresh, artisanal potato gnocchi	A039	3
Ñoquis de papa, envasados	Potato gnocchi, packaged	A140	3
Palmitos, enlatados	Palm hearts, canned	V029	3
Pan árabe	Pita bread (pan árabe)	A071	3
Pan cacho (tipo pan casero de campo, sin grasa)	Cacho bread (type of homemade country bread, without fat)	A215	3
Pan con grasa (tipo pan "cordero")	Bread with fat (type of "lamb" bread)	A240	3
Pan criollo / semita / tortillas / bollos / harinitas	Creole bread / semita / tortillas / rolls / flatbreads	A072	3
Pan de centeno	Rye bread	A040	3
Pan de gluten	Gluten bread	A042	3
Pan de maíz	Corn bread	A043	3
Pan de miga	Soft sandwich bread (pan de miga)	A044	3
Pan de salvado	Bran bread	A045	3
Pan de salvado con semillas	Bran bread with seeds	A184	3
Pan de salvado sin sal (panadería)	Unsalted bran bread (bakery)	A246	3
Pan de viena	Vienna bread	A046	3
Pan dulce	Sweet bread	A142	3
Pan figacita de manteca	Figacita butter bread	A169	3
Pan francés o pan casero sin agregado de grasa o aceite	French bread or homemade bread without added fat or oil	A041	3
Pan francés sin sal	French bread without salt	A245	3
Pan negro sin levadura	Black unleavened bread	A049	3
Pan rallado	Breadcrumbs	A050	3
Pan rallado, SIN TACC	Breadcrumbs, WITHOUT TACC	A226	3
Panceta	Bacon	C006	3
Panko (pan rallado japonés)	Panko (Japanese breadcrumbs)	A241	3
Pastas secas, SIN TACC	Dried pasta, WITHOUT TACC	A146	3
Pastelitos rellenos de dulce de membrillo o batata (panadería / confitería)	Pastries filled with quince or sweet potato sweets (bakery/confectionery)	A249	3
Pastrón	Pastrami	C058	3
Patay	Carob fruit bar (patay)	A147	3
Paté / picadillo	Pate / minced meat	C018	3
Pera, enlatada (fruta y almíbar)	Pear, canned (fruit and syrup)	F037	3
Pickles, en vinagre	Pickles, pickled	V070	3
Pionono	Sponge cake sheet (pionono)	A148	3
Pistacho salado	Salted pistachio	Z038	3
Pochoclo acaramelado	Caramel popcorn	A149	3
Pochoclo salado	Salty popcorn	A150	3
Porotos, enlatados	Beans, canned	A151	3
Quesillo (preparación regional, queso amasado artesanal)	Quesillo (regional preparation, artisanal kneaded cheese)	Q049	3
Queso Azul	Blue Cheese	Q022	3

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Queso blanco descremado, untable	Skim white cheese, spreadable	Q003	3
Queso blanco semidescremado, untable	Semi-skimmed white cheese, spreadable	Q002	3
Queso Cheddar	Cheddar	Q042	3
Queso Chubut	Chubut Cheese	Q023	3
Queso crema entero, untable	Whole cream cheese, spreadable	Q001	3
Queso Cremoso	Cream Cheese	Q050	3
Queso Cuartirolo	Cuartirolo Cheese	Q017	3
Queso de cabra	Goat cheese	Q045	3
Queso de cerdo	Headcheese	C024	3
Queso de máquina	Machine cheese	Q021	3
Queso de vaca criollo	Creole cow cheese	Q052	3
Queso en hebras	String cheese	Q034	3
Queso Fontina	Fontina Cheese	Q024	3
Queso fresco sin sal	Fresh unsalted cheese	Q047	3
Queso Goya	Goya Cheese	Q036	3
Queso Gruyére	Gruyére cheese	Q025	3
Queso Holanda	Holland cheese	Q026	3
Queso Limburguer	Limburger Cheese	Q018	3
Queso Mar del Plata	Mar del Plata-style cheese	Q027	3
Queso Mascarpone	Mascarpone Cheese	Q044	3
Queso Muzzarella	Mozzarella Cheese	Q019	3
Queso Parmesano	Parmesan Cheese	Q037	3
Queso pasta blanda descremado	Soft skim cheese	Q015	3
Queso pasta blanda doble crema	Double cream soft cheese	Q016	3
Queso pasta blanda, PROMEDIO (cuartirolo, doble crema, Limburguer, muzzarella)	Mozzarella)	Q006	3
Queso pasta dura, PROMEDIO (Goya, Parmesano, Provolone, Reggianito, Sardo, Sbrinz)	Sardo or Sbrinz cheese)	Q008	3
Queso pasta semidura (Holanda, Mar del Plata, Pategrás, Queso de máquina, Roquefort, Taif, Tandil)	Holland-type, Mar del Plata, Pategras, machine-pressed, Roquefort, Taif, Tandil cheeses)	Q007	3
Queso Pategrás	Pategras Cheese	Q028	3
Queso Provolone	Provolone	Q038	3
Queso rallado envasado	Packaged grated cheese	Q032	3
Queso Reggianito	Reggianito Cheese	Q039	3
Queso Roquefort	Roquefort cheese	Q029	3
Queso Sardo	Sardinian Cheese	Q040	3
Queso Sbrinz	Sbrinz cheese	Q041	3
Queso Taif	Taif cheese	Q030	3
Queso Tandil	Tandil cheese	Q031	3
Queso untable tipo Finlandia	Finnish spreadable cheese	Q013	3
Queso untable tipo Tholem	Tholem type spreadable cheese	Q011	3
Queso untable, PROMEDIO	Spreadable cheese, AVERAGE	Q000	3

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Quesos de pasta semidura descremados	Skim semi-hard cheeses	Q020	3
Quinoa inflada	Puffed Quinoa	A155	3
Rabas (restaurant)	Fried squid rings (restaurant)	P047	3
Ravioles frescos, artesanal	Fresh, artisanal ravioli	A054	3
Ravioles frescos, artesanal, SIN TACC	Fresh, artisanal ravioli, WITHOUT TACC	A225	3
Ricota	Ricotta	Q004	3
Ricota descremada	Skim ricotta	Q010	3
Rosca de pascuas (panadería / confitería)	Easter rosca (bakery / confectionery)	A223	3
Rosquete	Donut	A243	3
Rosquita tipo Doughtnut	Donut type donut	A210	3
Rosquito de vino (cuyano)	Wine donut (cuyano)	A209	3
Salame	Salami	C026	3
Salchicha parrillera	Grilled sausage	C028	3
Salchichón	Salchichón (dry-cured sausage)	C067	3
Sardinas, enlatadas al natural	Sardines, canned naturally	P044	3
Sardinas, enlatadas en aceite	Sardines, canned in oil	P013	3
Scones (panadería / confitería)	Scones (bakery/confectionery)	A196	3
Semilla de girasol con sal y con cáscara	Sunflower seed with salt and shell	Z049	3
Sidra	Cider	B015	3
Sorrentinos frescos, artesanal	Fresh, artisanal Sorrentinos	A200	3
Sushi, PROMEDIO	Sushi, AVERAGE	A183	3
Tapas para lasagna, pasta seca	Lasagna sheets, dry pasta	A222	3
Tomate, desecado	Tomato, dried	V068	3
Tomate, enlatado	Tomato, canned	V039	3
Tomate, puré de tomate	Tomato, tomato puree	V060	3
Torta de coco (panadería)	Coconut cake (bakery)	A238	3
Torta de manzana (panadería)	Apple pie (bakery)	A251	3
Torta de ricotta (panadería)	Ricotta cake (bakery)	A235	3
Torta frita	Fried dough (torta frita)	A062	3
Torta mil hojas	Mille-feuille cake	A228	3
Tortitas Jachaleras	Jachaleras Pancakes	A158	3
Tostadas de gluten	Gluten toast	A159	3
Tostadas de mesa	Table toast	A160	3
Tostadas integrales	Whole wheat toast	A161	3
Tostadas sin sal	Toast without salt	A163	3
Triguillo	Cracked wheat (bulgur)	A199	3
Turrón tipo navideño	Christmas type nougat	A174	3
Vainillas	Sponge fingers (vainillas)	A061	3
Vermouth	Vermouth	B016	3
Vino blanco	White wine	B017	3
Vino espumante, tipo Frizze	Sparkling wine, Frizze type	B120	3
Vino Mistela	Mistela Wine	B122	3
Vino rosado	Rosé wine	B123	3
Vino tinto	Red wine	B072	3

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
NOVA 4. Ultra-processed foods			
Aceto balsámico	Balsamic vinegar	O022	4
Aderezo Caesar o Ranch	Caesar or Ranch Dressing	O021	4
Aderezo para ensaladas light	Light salad dressing	O045	4
Aderezo Salsa Alioli	Aioli Sauce Dressing	O048	4
Aderezo Salsa Cheddar	Cheddar Sauce Dressing	O037	4
Adherezo deshidratado a base de queso Reggianito, sémola y almidones	Dehydrated dressing based on Reggianito cheese, semolina and starches	Q033	4
Agua saborizada Aquarius, varios sabores	Aquarius flavored water, various flavors	B044	4
Agua saborizada Awafrut, varios sabores	Awafrut flavored water, various flavors	B046	4
Agua saborizada con azúcar, PROMEDIO	Sugar Flavored Water, AVERAGE	B030	4
Agua saborizada Flip light	Flip light flavored water	B051	4
Agua saborizada H2O, varios sabores	H2O flavored water, various flavors	B049	4
Agua saborizada Levité Cero, varios sabores light	Levité Cero flavored water, various light flavors	B048	4
Agua saborizada Levité, varios sabores	Levité flavored water, various flavors	B045	4
Agua saborizada light, PROMEDIO	Light flavored water, AVERAGE	B031	4
Agua saborizada tipo Sierra de los Padres, Manaos, otros	Flavored water (Sierra de los Padres, Manaos, and similar brands)	B047	4
Agua saborizada We o Ser ligth	We or Ser light flavored water	B050	4
Alfajor de arroz	Rice-based alfajor	D037	4
Alfajor de chocolate	Chocolate-covered alfajor	D001	4
Alfajor de chocolate con relleno tipo mousse	Chocolate-covered alfajor with mousse filling	D028	4
Alfajor de dulce de leche	Alfajor with dulce de leche filling	D002	4
Alfajor de fruta	Fruit-filled alfajor	D036	4
Alfajor de maicena	Cornstarch alfajor	D029	4
Alfajor de miel	Honey alfajor	D107	4
Alfajor light	Low-calorie alfajor	D035	4
Alfajor santafesino, tapas cubiertas con glasé	Santa Fe-style alfajor, glazed layers	D104	4
Algodón de azúcar	Cotton candy	D118	4
Alimento a base de almendras (leche de almendras)	Almond-based food (almond milk)	Z043	4
Alimento a base de almendras con chocolate (chocolatada)	Food based on almonds with chocolate (chocolatada)	Z050	4
Alimento a base de castañas (leche de castañas)	Chestnut-based food (chestnut milk)	Z040	4
Alimento a base de coco (leche de coco, industrializada)	Coconut-based food (coconut milk, industrialized)	F069	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Alimento a base de leche entera y café, fortificado con calcio, varios sabores,	Food based on whole milk and coffee, fortified with calcium, various flavors,	L065	4
Amargo serrano light*	Serrano bitter light*	B002	4
Amargo serrano*	Serrano bitter*	B001	4
Ananá Fizz	Pineapple Fizz	B129	4
Ananá glaseado	Glazed pineapple	F071	4
Anchoa, pasta	Anchovy, pasta	P009	4
Arroz a la portuguesa (envasado, fortificado con vitaminas y minerales)	Portuguese-style rice (canned, fortified with vitamins and minerals)	A255	4
Arroz a la valenciana (envasado, fortificado con vitaminas y minerales)	Valencian rice (packaged, fortified with vitamins and minerals)	A257	4
Arroz con leche envasado listo para consumir	Ready-to-eat packaged rice pudding	D103	4
Arroz con leche envasado sabor caramelo, fortificado	Caramel flavor packaged rice pudding, fortified	D121	4
Arroz envasado deshidratado listo para preparar con salsa u otros ingredientes (tipo cuatro quesos, primavera, etc.)	(four cheese type, spring, etc.)	A103	4
Arroz inflado azucarado	Sweetened puffed rice	A191	4
Avena, almohaditas	Oatmeal, pillows	A104	4
Azúcar light	Light sugar	D070	4
Bananita con cobertura de chocolate	Banana with chocolate coating	D102	4
Baño de repostería para cubrir tortas	Pastry bath to cover cakes	D052	4
Barra crocante de arroz	Crispy rice bar	A105	4
Barra de arroz rellena bañada	Dipped stuffed rice bar	D041	4
Barra de cereal fortificada	Fortified cereal bar	A106	4
Barra de cereal light	Light cereal bar	A107	4
Barra de cereales, PROMEDIO	Cereal bar, AVERAGE	A008	4
Barra de chocolate tipo Kinder	Kinder chocolate bar	D053	4
Barra proteica	Protein bar	D132	4
Bebida a base de soja sabor natural	Natural flavor soy-based drink	B020	4
Bebida a base de soja y jugo, varios sabores	Soy and juice based drink, various flavors	B003	4
Bebida a base de soja y jugo, varios sabores light	Soy and juice-based drink, various light flavors	B004	4
Bebida energizante	Energy drink	B041	4
Bebida láctea a base de agua, yogurt, azúcar y saborizantes	Dairy drink based on water, yogurt, sugar and flavorings	Y019	4
Bebida láctea parcialmente descremada fluida, baja en lactosa, fortificada con vitaminas A, D, B2, B9 y zinc	Vitamins A, D, B2, B9 and zinc	L068	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Bebida mezcla de fernet con gaseosa sabor cola, lista para consumir	Drink mix of fernet with cola flavor soda, ready to consume	B113	4
Bebidas destiladas, PROMEDIO	Distilled beverages, AVERAGE	B005	4
Bizcochitos de arroz envasados (tipo galletita)	Packaged rice cakes (cracker type)	A110	4
Bizcochitos de grasa envasados	Packaged Fat Biscuits	A108	4
Bizcochitos de grasa envasados light	Light packaged fat biscuits	A109	4
Bizcochuelo preparado (panadería)	Prepared sponge cake (bakery)	A229	4
Bizcochuelo preparado a partir de polvo premezcla	Sponge cake prepared from premix powder	A111	4
Bizcochuelo preparado a partir de polvo premezcla, SIN TACC	Sponge cake prepared from premix powder, WITHOUT TACC	A112	4
Bocadito con dulce de leche tipo Cabsha	Bite-size chocolate with dulce de leche (Cabsha-style)	D054	4
Bocadito tipo Dos Corazones	Two Hearts Snack	D113	4
Bocadito tipo Marroc	Moroccan snack	D115	4
Bocaditos de pescado, prefritos	Fish bites, pre-fried	P041	4
Bombón de chocolate relleno de licor	Chocolate bonbon filled with liquor	D129	4
Bombón helado	Frozen bonbon	D066	4
Bombón tipo Bon O Bon	Bon O Bon type chocolate	D055	4
Bombones de fruta	Fruit bonbons	D044	4
Bondiola (fiambre)	Bondiola (deli meat)	C094	4
Budín industrializado	Industrialized pudding	A114	4
Budín industrializado con nueces, tipo navideño	Industrialized pudding with nuts, Christmas type	A234	4
Cacao en polvo tipo Nesquik*	Nesquik type cocoa powder*	D051	4
Cacao en polvo tipo Zucoa / Zucoa Max instantáneo otros sabores*	Zucoa type cocoa powder / Zucoa Max instant other flavors*	D108	4
Cacao, cascarilla*	Cocoa, husk*	D117	4
Café con leche en cápsula*	Coffee with milk in capsule*	B117	4
Café instantáneo (polvo o granulado con leche y azúcar para preparar)*	Instant coffee (powder or granules with milk and sugar to prepare)*	B060	4
Café instantáneo (polvo o granulado para preparar)*	Instant coffee (powder or granules for brewing)*	B057	4
Café instantáneo descafeinado (polvo o granulado para preparar)*	Instant decaffeinated coffee (powder or granules for brewing)*	B056	4
Café molido en cápsula*	Ground coffee in capsule*	B059	4
Caldos en cubo*	Broth cubes*	O007	4
Caldos en polvo light*	Broth powder light*	O008	4
Caldos en polvo sin sal light*	Broth powder without light salt*	O009	4
Caldos en polvo*	Broth powder*	O039	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Cantimpalo (fiambre)	Cantimpalo (cold meat)	C121	4
Capelettis / capelettinis deshidratados, envasados	Dehydrated capelettis / capelettinis, packaged	A115	4
Capucchino instantáneo (polvo o granulado para preparar light)*	Instant cappuccino (powder or granules to prepare light)*	B062	4
Capucchino instantáneo (polvo o granulado para preparar)*	Instant cappuccino (powder or granules to prepare)*	B063	4
Capucchino preparado	Prepared cappuccino	B061	4
Caramelos duros	Hard candies	D014	4
Caramelos duros light	Light hard candies	D015	4
Caramelos masticables	Chewy candies	D013	4
Caramelos rellenos	Filled candies	D012	4
Cereal desayuno, aritos frutales	Breakfast cereal, fruit rings	A074	4
Cereal desayuno, aritos sabor miel	Breakfast cereal, honey flavor rings	A182	4
Cereal desayuno, bolitas de chocolate	Breakfast cereal, chocolate balls	A194	4
Cereal desayuno, copos azucarados, fortificados	Breakfast cereal, sugared flakes, fortified	A117	4
Cereal desayuno, copos azucarados, sin fortificar	Breakfast cereal, sugar flakes, unfortified	A012	4
Cereal desayuno, copos de cereal integral (arroz y maíz), fortificados	Breakfast cereal, whole grain cereal flakes (rice and corn), fortified	A180	4
Cereal desayuno, copos de maíz sin azúcar, sin fortificar	Breakfast cereal, unsweetened cornflakes, unfortified	A075	4
Cereal desayuno, salvado de trigo tipo bastoncitos	Breakfast cereal, wheat bran sticks	A057	4
Cereal infantil para papilla	Children's cereal for porridge	A116	4
Chicles	Chewing gum	D042	4
Chicles sin azúcar	Sugar-free chewing gum	D043	4
Chipá preparado a partir de polvo premezcla	Chipá prepared from premix powder	A120	4
Chips de chocolate	Chocolate chips	D123	4
Chocolate 60% cacao o más	Cocoa or more	D057	4
Chocolate amargo	Dark chocolate	D126	4
Chocolate blanco	White chocolate	D058	4
Chocolate con almendras	Chocolate with almonds	D059	4
Chocolate con maní	Chocolate with peanuts	D056	4
Chocolate light	Light chocolate	D060	4
Chocolate para taza	Chocolate for cup	D016	4
Chocolate relleno con pasta de maní tipo Snickers	Chocolate filled with Snickers-type peanut paste	D119	4
Chocolate relleno de dulce de leche tipo Tofi	Chocolate filled with dulce de leche (Tofi-style)	D095	4
Chocolatín / chocolate con leche	Chocolate / milk chocolate	D017	4
Chupetín	Lollipop	D047	4
Conito de dulce de leche envasado	Packaged dulce de leche cone confection	D034	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Crema chantilly (lista para consumir, aerosol o industrial preparada)	Chantilly cream (ready to consume, aerosol or industrial prepared)	G018	4
Crema de avellanas con cacao tipo Nutella	Hazelnut cream with Nutella-type cocoa	D114	4
Crema light	Light cream	G016	4
Crema saborizada envasada tipo Ducrem	Ducrem-type packaged flavored cream	D110	4
Cubanito relleno	Stuffed Cubanito	D033	4
Dr. Lemon	Dr Lemon	B115	4
Dulce de batata light	Light sweet potato paste	D081	4
Dulce de leche	Dulce de leche	D008	4
Dulce de leche light	Light dulce de leche	D077	4
Dulce de membrillo light	Light quince paste	D082	4
Durazno, enlatado light (fruta y almíbar)	Peach, light canned (fruit and syrup)	F013	4
Edulcorante en polvo con aspartamo y acesulfame k	Powdered sweetener with aspartame and acesulfame k	D083	4
Edulcorante en polvo con sacarina y ciclamato (tipo Hileret clásico 1 a 10, Sucaryl	Powdered sweetener with saccharin and cyclamate (classic Hileret type 1 to 10, Sucaryl	D084	4
Edulcorante en polvo con stevia	Powdered sweetener with stevia	D085	4
Edulcorante en polvo con sucralosa	Powdered sweetener with sucralose	D086	4
Edulcorante en polvo, PROMEDIO	Powdered Sweetener, AVERAGE	D071	4
Edulcorante en tabletas o comprimidos con aspartamo y acesulfame k	Sweetener in tablets or tablets with aspartame and acesulfame k	D093	4
Edulcorante en tabletas o comprimidos con ciclamato y sacarina	Sweetener in tablets or tablets with cyclamate and saccharin	D094	4
Edulcorante en tabletas o comprimidos sin identificar	Unidentified sweetener in tablets or tablets	D092	4
Edulcorante líquido con aspartamo	Liquid sweetener with aspartame	D088	4
Edulcorante líquido con ciclamato, sacarina y sucralosa	Liquid sweetener with cyclamate, saccharin and sucralose	D091	4
Edulcorante líquido con sacarina y ciclamato	Liquid sweetener with saccharin and cyclamate	D089	4
Edulcorante líquido con stevia	Liquid sweetener with stevia	D072	4
Edulcorante líquido con sucralosa	Liquid sweetener with sucralose	D090	4
Edulcorante líquido, PROMEDIO	Liquid Sweetener, AVERAGE	D087	4
Farofa tradicional, harina de mandioca condimentada, envasada lista para	Traditional farofa, seasoned cassava flour, packaged ready for	A192	4
Fernet	Fernet bitter spirit	B043	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Fiambre de cerdo	Pork cold cuts	C051	4
Flan envasado listo para consumir	Ready-to-eat packaged flan	Y006	4
Flan envasado listo para consumir light	Packaged flan ready to consume light	Y007	4
Fondant	Fondant icing	D130	4
Frutas abrillantadas	Candied fruits	F065	4
Galleta de campo	Country-style cracker	A064	4
Galleta de grasa con chicharrón	Lard cracker with pork cracklings	A206	4
Galleta marinera	Ship's biscuit (hardtack)	A065	4
Galleta marinera con semillas / galletitas crackers con semillas	Ship's biscuit with seeds / seeded crackers	A188	4
Galleta marinera integral	Wholegrain ship's biscuit (hardtack)	A253	4
Galleta marinera, light en grasas	Ship's biscuit, light in fat	A203	4
Galletas de arroz	Rice crackers	A019	4
Galletas de arroz horneadas tipo snack varios sabores	Baked snack-type rice crackers various flavors	A185	4
Galletas de arroz sin sal	Unsalted rice crackers	A123	4
Galletitas de agua con girasol alto oleico	Water cookies with high oleic sunflower	A127	4
Galletitas de agua con grasa vacuna	Water biscuits with beef fat	A021	4
Galletitas de agua light	Light water cookies	A247	4
Galletitas de agua sin sal	Unsalted water crackers	A066	4
Galletitas de agua sin sal con chía y lino	Unsalted water cookies with chia and flax	A258	4
Galletitas de agua, PROMEDIO	Water Cookies, AVERAGE	A020	4
Galletitas de salvado o integrales	Bran or whole wheat crackers	A022	4
Galletitas de vainilla con chips de chocolate (panadería)	Vanilla cookies with chocolate chips (bakery)	A254	4
Galletitas dulces con centro de membrillo o similar (tipo Pepas)	Sweet cookies with a quince center or similar (Pepas type)	A186	4
Galletitas dulces con chips de chocolate	Sweet cookies with chocolate chips	A181	4
Galletitas dulces de almidón de maíz (tapita de alfajor)	Sweet cornstarch cookies (alfajor shells)	A177	4
Galletitas dulces hojaldradas	Sweet flaky cookies	A220	4
Galletitas dulces obleas	Sweet wafer cookies	A068	4
Galletitas dulces rellenas	Stuffed sweet cookies	A023	4
Galletitas dulces rellenas bañadas en chocolate	Stuffed sweet cookies dipped in chocolate	A211	4
Galletitas dulces simples	Simple sweet cookies	A024	4
Galletitas dulces sin azúcar agregada, con edulcorante (dietética)	Sweet cookies without added sugar, with sweetener (dietary)	A205	4
Galletitas dulces tipo Frutigran	Frutigran type sweet cookies	A125	4
Galletitas dulces, SIN TACC	Sweet cookies, WITHOUT TACC	A124	4
Galletitas integrales sin sal	Whole grain crackers without salt	A067	4
Galletitas tipo copetín	Cupcake type cookies	A126	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Galletitas tipo talitas o fajitas	Talitas or fajita type cookies	A198	4
Garrapiñada de maní	Peanut candy	D080	4
Gaseosa con azúcar, agua tónica	Soda with sugar, tonic water	B025	4
Gaseosa con azúcar, PROMEDIO	Soda with sugar, AVERAGE	B008	4
Gaseosa con azúcar, sabor cola	Soda with sugar, cola flavor	B021	4
Gaseosa con azúcar, sabor lima limón	Soda with sugar, lemon-lime flavor	B022	4
Gaseosa con azúcar, sabor naranja	Soda with sugar, orange flavor	B023	4
Gaseosa con azúcar, sabor pomelo	Soda with sugar, grapefruit flavor	B024	4
Gaseosa light, Guaraná	Light soda, Guarana	B112	4
Gaseosa light, PROMEDIO	Light soda, AVERAGE	B009	4
Gaseosa light, sabor cola	Light soda, cola flavor	B026	4
Gaseosa light, sabor lima limón	Light soda, lemon-lime flavor	B027	4
Gaseosa light, sabor naranja	Light soda, orange flavor	B028	4
Gaseosa light, sabor pomelo	Light soda, grapefruit flavor	B029	4
Gatorade Zero, varios sabores (sin azúcar)	Gatorade Zero, various flavors (sugar-free)	B039	4
Gatorade, varios sabores	Gatorade, various flavors	B037	4
Gelatina envasada lista para consumir	Packaged gelatin ready to consume	D069	4
Gelatina light, preparada a partir de polvo	Light gelatin, prepared from powder	D022	4
Gelatina preparada a partir de polvo	Gelatin prepared from powder	D021	4
Golosinas de merengue y cucurucho tipo Heladitos / Gallinitas	Meringue and ice cream cone type treats / Gallinitas	D125	4
Gomitas de gelatina	Gelatin gummies	D048	4
Grana para decorar tortas	Grana to decorate cakes	D098	4
Granadina Zero (sin azúcar)	Grenadine Zero (sugar free)	B131	4
Granola	Granola	A026	4
Guiso de lentejas (envasado, fortificado con vitaminas y minerales)	Lentil stew (canned, fortified with vitamins and minerals)	A244	4
Habanitos de chocolate	Chocolate-coated habanitos	D122	4
Hamburguesa de carne vacuna (carnicería)	Beef burger (butcher shop)	C118	4
Hamburguesa de carne vacuna, industrializada	Beef burger, industrialized	C011	4
Hamburguesa de carne vacuna, industrializada light	Beef burger, light industrialized	C053	4
Hamburguesa de carne vacuna, industrializada, rebozada	Beef burger, industrialized, battered	C124	4
Hamburguesa de cerdo (carnicería)	Pork burger (butcher shop)	C116	4
Hamburguesa de pollo (pollería / granja)	Chicken burger (poultry / farm)	C113	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Hamburguesa de pollo, industrializada	Chicken burger, industrialized	C010	4
Hamburguesa de quinoa envasada	Packaged quinoa burger	A260	4
Hamburguesa de soja congelada	Frozen soy burger	A175	4
Harina de trigo con levadura para pizza	Wheat flour with yeast for pizza	A202	4
Helado de agua (heladería)	Water ice / sorbet (ice cream shop)	D064	4
Helado de agua (palito)	Ice pop / popsicle	D019	4
Helado de crema (heladería)	Cream ice cream (ice cream shop)	D020	4
Helado de crema envasado	Packaged ice cream	D065	4
Helado envasado con cucurucho, PROMEDIO	Ice cream packaged with cone, AVERAGE	D067	4
Huevo de chocolate tipo Kinder	Kinder Chocolate Egg	D061	4
Jamón cocido	Cooked ham	C004	4
Jugo concentrado con azúcar para diluir*	Concentrated juice with sugar to dilute*	B014	4
Jugo concentrado para diluir light*	Concentrated juice to dilute light*	B118	4
Jugo congelado con azúcar en sachet	Frozen juice with sugar in sachet	B054	4
Jugo de arándanos y chía	Cranberry and chia juice	B078	4
Jugo de fruta envasado varios sabores	Packaged fruit juice various flavors	B010	4
Jugo de limón, envasado	Lemon juice, packaged	B069	4
Jugo en polvo BC*	BC Juice Powder*	B052	4
Jugo en polvo con azúcar*	Powdered juice with sugar*	B053	4
Jugo en polvo con azúcar, PROMEDIO*	Juice powder with sugar, AVERAGE*	B011	4
Jugo en polvo light*	Light powdered juice*	B012	4
Jugo tipo BC, varios sabores (alimento líquido dietético con 50% de jugo y pulpa)	Of juice and pulp)	B121	4
Kani Kama	Imitation crab sticks (kani kama)	P039	4
Ketchup	Ketchup (tomato sauce)	O005	4
Leche chocolatada en polvo, fortificada, para desayunos escolares	Chocolate milk powder, fortified, for school breakfasts*	L044	4
Leche descremada en polvo, tipo Nido Buen Día, fortificada con hierro y	Skimmed milk powder, Nido Buen Día type, fortified with iron and	L069	4
Leche entera en polvo tipo Nido Fortigrow*	Nido Fortigrow* whole milk powder	L055	4
Leche entera fluida, con CLA, fortificada con vitaminas A y D	Fluid whole milk, with CLA, fortified with vitamins A and D	L066	4
Leche fórmula 0-12 meses, hipoalergénica, tipo Nutrilón HA, NAN HA, en	Formula milk 0-12 months, hypoallergenic, type Nutrilón HA, NAN HA, in	L063	4
Leche fórmula 0-12 meses, sin lactosa, en polvo*	Milk formula 0-12 months, lactose-free, powder*	L062	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Leche fórmula antireflujo, en polvo*	Anti-reflux formula milk, powder*	L042	4
Leche fórmula etapa 3, en polvo*	Stage 3 formula milk, powder*	L035	4
Leche fórmula etapa 3, fluida	Milk formula stage 3, fluid	L036	4
Leche fórmula etapa 3, sin lactosa, en polvo*	Milk formula stage 3, lactose-free, powder*	L061	4
Leche fórmula etapa 4, en polvo*	Stage 4 formula milk, powder*	L037	4
Leche fórmula etapa 4, fluida	Milk formula stage 4, fluid	L038	4
Leche fórmula inicio, en polvo*	Starter formula milk, powder*	L022	4
Leche fórmula inicio, fluida	Starter formula milk, fluid	L024	4
Leche fórmula para prematuros etapa 1, en polvo*	Premature formula, stage 1, powder*	L049	4
Leche fórmula para prematuros etapa 1, fluida	Premature formula stage 1, fluid	L050	4
Leche fórmula para prematuros etapa 2, en polvo*	Stage 2 preterm formula, powder*	L051	4
Leche fórmula para prematuros etapa 2, fluida	Premature formula milk stage 2, fluid	L052	4
Leche fórmula seguimiento, en polvo*	Follow-up formula milk, powder*	L023	4
Leche fórmula seguimiento, fluida	Follow-up formula milk, fluid	L025	4
Leche fórmula, tipo Alfaré, con proteína extensamente hidrolizada, triglicéridos de cadena media, DHA/GLA y sin lactosa, para alergia a la proteína de leche de vaca y problemas de malabsorción intestinal, en polvo*	Medium chain triglycerides, DHA/GLA and lactose-free, for allergy to	L057	4
Leche fórmula, tipo Enfabebe Confort, en polvo*	Formula milk, Enfabebe Confort type, powder*	L059	4
Leche fórmula, tipo GA1 Anamix Infant, en polvo*	Formula milk, type GA1 Anamix Infant, powder*	L056	4
Leche fórmula, tipo Nutrilon Pepti junior HE, en polvo*	Formula milk, type Nutrilon Pepti junior HE, powder*	L058	4
Leche parcialmente descremada fluida, con Fitoesteroles y ác. Grasos Omega 3, fortificada con calcio y vitaminas A, D y E	3, fortified with calcium and vitamins A, D and E	L053	4
Leche parcialmente descremada, sabor chocolate o dulce de leche, fortificada con vitaminas A y D, lista para consumir	Fortified reduced-fat flavored milk (chocolate or dulce de leche)	L001	4
Leche seguimiento etapa 3, sabor vainilla	Stage 3 monitoring milk, vanilla flavor	L064	4
Lentejas de chocolate	Chocolate lentil candies (M&M-style)	D063	4
Licor, PROMEDIO	Liquor, AVERAGE	B013	4
Madalenas	Muffins	A135	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Madalenas fortificadas	Fortified muffins	A136	4
Madalenas rellenas	Stuffed muffins	A187	4
Malta instantánea polvo*	Instant malt powder*	B130	4
Malvaviscos	Marshmallows	D101	4
Maní con chocolate	Peanuts with chocolate	D062	4
Maní japonés	Japanese-style coated peanuts	O043	4
Manteca light	Light butter	G015	4
Mantequilla de maní	Peanut butter	Z048	4
Margarina (en pote y en pan)	Margarine (tub and bar)	G011	4
Margarina untable light	Light spreadable margarine	G012	4
Masa de pizza, preparada a partir de polvo premezcla	Pizza dough, prepared from premix powder	A197	4
Masa de pizza, preparada a partir de polvo premezcla, SIN TACC	Pizza dough, prepared from premix powder, TACC FREE	A138	4
Masa de prepizza de panadería o envasada	Bakery or packaged prepizza dough	A038	4
Masa de tarta o empanadas light	Cake dough or light empanadas	A139	4
Matambre (fiambre)	Matambre (cold meat)	C057	4
Mayonesa	Mayonnaise	O001	4
Mayonesa de soja	Soy mayonnaise	O003	4
Mayonesa light	Light mayonnaise	O002	4
Mc Donald's, Cajita Feliz, Mc Nuggets X4 (ingresar cantidad de porciones)	S, Happy Meal, Mc Nuggets X4 (enter number of servings)	R040	4
Mc Donald's, Cajita Feliz, Mc Tostado (ingresar cantidad de porciones)	S, Happy Meal, Mc Tostado (enter number of servings)	R043	4
Mc Donald's, Cajita Feliz, Papas Kids (ingresar cantidad de porciones)	S, Happy Meal, Papas Kids (enter number of servings)	R041	4
Mc Donald's, Cajita Feliz, Papas pequeñas (ingresar cantidad de porciones)	S, Happy Meal, Small Potatoes (enter number of servings)	R042	4
Mc Donald's, Club House (ingresar cantidad de porciones)	S, Club House (enter number of servings)	R054	4
Mc Donald's, Clubhouse/Guacamole/Cryspi Onion BBQ con pollo crispy (ingresar cantidad de porciones)	S, Clubhouse/Guacamole/Cryspi Onion BBQ with crispy chicken (enter	R058	4
Mc Donald's, Clubhouse/Guacamole/Cryspi Onion BBQ con pollo grill (ingresar cantidad de porciones)	S, Clubhouse/Guacamole/Cryspi Onion BBQ with grilled chicken (enter	R059	4
Mc Donald's, Cono (ingresar cantidad de porciones)	S, Cone (enter number of servings)	R064	4
Mc Donald's, Cuarto de Libra (ingresar cantidad de porciones)	S, Quarter Pounder (enter number of servings)	R052	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Mc Donald's, Doble Cuarto de Libra con queso (ingresar cantidad de porciones)	S, Double Quarter Pounder with cheese (enter number of servings)	R049	4
Mc Donald's, Hamburguesa (ingresar cantidad de porciones)	S, Hamburger (enter number of servings)	R047	4
Mc Donald's, Hamburguesa con queso (ingresar cantidad de porciones)	S, Cheeseburger (enter number of servings)	R048	4
Mc Donald's, Mc Flurry Oreo (ingresar cantidad de porciones)	S, Mc Flurry Oreo (enter number of servings)	R068	4
Mc Donald's, Mcnífica (ingresar cantidad de porciones)	S, Mcnifica (enter number of servings)	R050	4
Mc Donald's, McNuggets x 10 (ingresar cantidad de porciones)	S, McNuggets x 10 (enter number of servings)	R056	4
Mc Donald's, McPollo Grill (ingresar cantidad de porciones)	S, McPollo Grill (enter number of servings)	R057	4
Mc Donald's, Papas grandes (ingresar cantidad de porciones)	S, Large potatoes (enter number of servings)	R063	4
Mc Donald's, Papas medianas (ingresar cantidad de porciones)	S, Medium Potatoes (enter number of servings)	R062	4
Mc Donald's, Papas pequeñas (ingresar cantidad de porciones)	S, Small potatoes (enter number of servings)	R061	4
Mc Donald's, Pollo Jr. (ingresar cantidad de porciones)	S, Chicken Jr. (enter number of servings)	R060	4
Mc Donald's, Sundae chocolate (ingresar cantidad de porciones)	S, Chocolate Sundae (enter number of servings)	R065	4
Mc Donald's, Sundae dulce de leche (ingresar cantidad de porciones)	Sundae with dulce de leche (enter number of servings)	R066	4
Mc Donald's, Sundae frutilla (ingresar cantidad de porciones)	S, Strawberry Sundae (enter number of servings)	R067	4
Mc Donald's, Triple Mac (ingresar cantidad de porciones)	S, Triple Mac (enter number of servings)	R051	4
McDonald's, BigMc (ingresar cantidad de porciones)	McDonald's, BigMc (enter number of servings)	R053	4
McDonald's, Cajita Feliz, Hamburguesa con queso Grill (ingresar cantidad de	McDonald's, Happy Meal, Grill Cheeseburger (enter quantity of	R046	4
McDonald's, Cajita Feliz, Hamburguesa Grill (ingresar cantidad de porciones)	McDonald's, Happy Meal, Burger Grill (enter number of servings)	R045	4
Medallón de menta y chocolate	Mint and chocolate medallion	D131	4
Medallón de pescado relleno con queso y espinaca, prefrito	Fish medallion stuffed with cheese and spinach, pre-fried	P043	4
Medallón de pollo relleno de jamón y queso, prefrito congelado	Chicken medallion stuffed with ham and cheese, pre-fried frozen	C114	4
Medallones de verdura congelados industrializados	Industrialized frozen vegetable medallions	V072	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Merengues / merenguitos	Meringues / meringues	D040	4
Mermelada de frutas	Fruit jam	D010	4
Mermelada de frutas light	Light fruit jam	D025	4
Mezcla de semillas (sésamo, amapola, amaranto) y almendras o similar, fortificado con calcio	Fortified with calcium	Z047	4
Mielcitas	Honey-flavored candies (mielcitas)	D106	4
Milanesa de arroz, prefrita	Rice Milanese, pre-fried	A165	4
Milanesa de lentejas (dietética)	Lentil Milanese (diet)	A259	4
Milanesa de pescado, prefrita	Fish Milanese, pre-fried	P040	4
Milanesa de pollo, prefrita congelada	Chicken Milanese, frozen pre-fried	C069	4
Milanesa de soja, prefrita	Soy Milanese, pre-fried	A164	4
Minitorta bañada	Dipped mini cake	D038	4
Mortadela	Mortadella	C017	4
Mostaza	Mustard	O006	4
Muffins veganos	Vegan muffins	A217	4
Muzzarella rebozada prefrita	Pre-fried breaded mozzarella	Q043	4
Nachos	Tortilla chips (nachos)	O028	4
Ñoquis de espinaca envasados	Packaged spinach gnocchi	A233	4
Ñoquis preparados a partir de polvo premezcla	Gnocchi prepared from premix powder	A171	4
Ñoquis preparados a partir de polvo premezcla, SIN TACC	Gnocchi prepared from premix powder, TACC FREE	A172	4
Nuggets o patitas de pollo	Chicken nuggets or fingers	C070	4
Oblea bañada tipo Kit Kat	Kit Kat type dipped wafer	D032	4
Oblea bañada tipo Nugaton	Nugaton type dipped wafer	D039	4
Paleta (fiambre)	Shoulder (cold meat)	C097	4
Palitos aireados de harina de maíz o de colores	Airy corn flour or colored sticks	O013	4
Palitos salados	Salty sticks	O014	4
Pan blanco, tipo molde, lacteado	White bread, mold type, dairy	A047	4
Pan de molde con salvado	Sliced bread with bran	A048	4
Pan de molde con salvado sin sal	Sliced bread with bran without salt	A141	4
Pan lactal, SIN TACC	Lactal bread, WITHOUT TACC	A237	4
Pan para panchos o hamburguesas, envasado	Bread for hot dogs or hamburgers, packaged	A143	4
Pan paraguayo / galleta paraguaya (bollitos con levadura, harina, grasa o	Paraguayan bread / Paraguayan cookie (rolls with yeast, flour, fat or	A231	4
Pan rallado, fortificado	Breadcrumbs, fortified	A144	4
Papa, croquetas u otro preformado de papa rebozado, prefrito, congelado	Potatoes, croquettes or other preformed potato in batter, pre-fried, frozen	A153	4
Papa, puré instantáneo (polvo)*	Potato, instant mashed (powder)*	A152	4
Papas fritas de copetín	Potato chips (crisps)	O015	4
Papas prefritas congeladas envasadas	Packaged frozen pre-fried potatoes	A145	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Pasas de uva con chocolate	Raisins with chocolate	D133	4
Pastillas	Pastilles (hard candy)	D045	4
Pastillas sin azúcar	Sugar-free pastilles	D046	4
Pera, enlatada light (fruta y almíbar)	Pear, canned light (fruit and syrup)	F038	4
Polvo para preparar brownie*	Powder to prepare brownie*	A189	4
Polvo para preparar flan*	Powder to prepare flan*	D009	4
Polvo para preparar helado*	Powder to prepare ice cream*	D068	4
Polvo para preparar mousse de limón / chocolate*	Powder to prepare lemon / chocolate mousse*	D112	4
Polvo para preparar postre lácteo light*	Powder to prepare light dairy dessert*	D076	4
Polvo para preparar postre lácteo*	Powder to prepare dairy dessert*	D075	4
Postre a base de pasta de maní, tipo Mantecol	Dessert based on peanut paste, Mantecol type	D049	4
Postre de leche bebible infantil	Children's drinkable milk dessert	L013	4
Postre de leche listo para consumir light	Milk dessert ready to consume light	L015	4
Postre de leche listo para consumir tipo Danonino	Ready-to-eat milk dessert Danonino type	L016	4
Postre de leche listo para consumir tipo Serenito	Milk dessert ready to consume Serenito type	L014	4
Powerade Zero, varios sabores (sin azúcar)	Powerade Zero, various flavors (sugar-free)	B040	4
Powerade, varios sabores	Powerade, various flavors	B038	4
Prefritos congelados, croquetas de vegetales (espinaca, brócoli)	Frozen pre-fried, vegetable croquettes (spinach, broccoli)	V062	4
Prefritos congelados, milanesas de vegetales (4 vegetales, espinaca)	Frozen pre-fried, vegetable milanesas (4 vegetables, spinach)	V063	4
Prefritos congelados, patitas de vegetales (espinaca)	Frozen pre-fried vegetables (spinach)	V064	4
Premezcla para bizcochuelo, SIN TACC	Premix for sponge cake, WITHOUT TACC	A256	4
Premezcla para panificados, pastas y postres, SIN TACC	Premix for baked goods, pastas and desserts, WITHOUT TACC	A213	4
Probióticos descremados	Skim probiotic dairy drink	L017	4
Probióticos enteros	Whole-fat probiotic dairy drink	L018	4
Puflito	Puffito	A154	4
Pulpa de frutas para repostería, coberturas y rellenos (frutilla, arándano o maracuyá)	Fruit pulp for baking, toppings and fillings (strawberry, blueberry or passion fruit)	D099	4
Puré de frutas envasado (alimento infantil)	Packaged fruit puree (baby food)	F057	4
Queso en hebras light	Light string cheese	Q035	4
Queso Fontina fundido light	Light melted Fontina cheese	Q051	4
Queso fundido	Processed cheese	Q005	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Queso porsalut sin sal light	Porsalut cheese without light salt	Q046	4
Queso untable tipo Finlandia light	Finland light spreadable cheese	Q014	4
Queso untable tipo Tholem light	Tholem light spreadable cheese	Q012	4
Ravioles deshidratados, envasados	Dehydrated ravioli, packaged	A157	4
Ravioles frescos, envasados	Fresh ravioli, packaged	A156	4
Relleno y cobertura para repostería lista para batir, varios sabores	Filling and topping for ready-to-mix pastries, various flavors	D109	4
Rhodesia	Rhodesia chocolate biscuit bar (brand-style)	D030	4
Rocío vegetal	Vegetable cooking spray	Z024	4
Rocío vegetal, para cocción	Vegetable cooking spray, for cooking	Z037	4
Rosquita tipo Doughtnut bañada en chocolate	Donut type donut dipped in chocolate	A227	4
Rosquita tipo Doughtnut de chocolate	Chocolate donut type donut	A219	4
Saborizador en cubos	Cubed flavoring	O024	4
Saborizador en polvo	Powder flavoring	O038	4
Saborizadores Tecknofood, sabores varios (malta, vainilla)	Tecknofood flavorings, various flavors (malt, vanilla)	D124	4
Sal dietética o modificada, PROMEDIO	Dietary or modified salt, AVERAGE	O033	4
Sal dietética, sin sodio	Diet salt, sodium-free	O027	4
Salchicha de viena	Vienna sausage	C027	4
Salchicha de viena light	Light Vienna sausage	C066	4
Salchicha rellena con cheddar	Cheddar Stuffed Sausage	C119	4
Salsa a base de tomate en tetra brick o sachet, lista para consumir (ej: pomarola,	Tomato-based sauce in tetra brick or sachet, ready to consume (e.g. pomarola,	O035	4
Salsa agridulce	Sweet and sour sauce	O047	4
Salsa barbacoa	Barbecue sauce	O020	4
Salsa blanca en polvo envasada*	Packaged powdered white sauce*	O049	4
Salsa chimichurri envasada	Packaged chimichurri sauce	O051	4
Salsa de soja	Soy sauce	O034	4
Salsa deshidratada varios sabores, polvo*	Dehydrated sauce various flavors, powder*	O046	4
Salsa Golf	Golf sauce (pink sauce)	O004	4
Salsa para guiso campero envasada, fortificada*	Sauce for packaged country stew, fortified*	O040	4
Salsa topping de chocolate	Chocolate topping sauce	D097	4
Salsa topping de dulce de leche	Dulce de leche topping sauce (dessert syrup)	D096	4
Seitán, sucedáneo de carne a base de gluten de trigo	Seitan, meat substitute based on wheat gluten	A214	4
Snacks saborizados salados a base de maíz	Corn-based salty flavored snacks	O041	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Snacks saborizados salados a base de sémola de arroz, horneados, tipo Palitos de arroz / aros de cebolla	Salty flavored snacks based on rice semolina, baked, type of sticks	O044	4
Soja texturizada	Textured soy	A170	4
Sopa crema instantánea preparada	Ready-made instant cream soup	O010	4
Sopa crema instantánea, polvo para preparar (sabores varios)*	Instant cream soup, preparation powder (various flavors)*	O029	4
Sopa crema instantánea, polvo para preparar light (sabores varios)*	Instant cream soup, powder to prepare light (various flavors)*	O030	4
Sopa crema preparada a partir de polvo premezcla light	Cream soup prepared from light premix powder	O011	4
Sopa crema preparada a partir de polvo premezcla sin sal light	Cream soup prepared from premix powder without light salt	O012	4
Sopa de fideos deshidratada envasada para preparar*	Packaged Dehydrated Noodle Soup to Prepare*	O025	4
Sopa de vegetales instantánea, polvo para preparar*	Instant Vegetable Soup, Brewing Powder*	O036	4
Suprema de pollo con espinaca y queso, prefrita congelada	Chicken supreme with spinach and cheese, frozen pre-fried	C071	4
Suprema de pollo con jamón y queso, prefrita congelada	Chicken supreme with ham and cheese, frozen pre-fried	C072	4
Tableta de dulce de leche	Dulce de leche fudge bar	D078	4
Tableta de dulce de leche light	Sugar-free dulce de leche bar	D079	4
Té concentrado de hierbas Herbalife	Herbalife Herbal Concentrate Tea	B132	4
Té saborizado en saquitos, varios sabores (ingresar cantidad de saquitos)*	Flavored tea in bags, various flavors (enter number of bags)*	B077	4
Tita	Tita chocolate biscuit bar (brand-style)	D031	4
Torta de cumpleaños (masa, relleno y cobertura). Comprada lista para consumir,	Birthday cake (dough, filling and topping). Purchased ready to consume,	A224	4
Torta helada envasada (crema helada varios sabores con base de galleta)	Packaged ice cream cake (various flavors ice cream with cookie base)	D111	4
Tortillas de trigo, envasadas	Wheat tortillas, packaged	A208	4
Tostadas light	Light toast	A162	4
Tostaditas saladas, saborizadas, producto tipo copetín	Salty, flavored toasts, snack-type product	A221	4
Trigo inflado azucarado	Sweetened puffed wheat	A230	4
Turrón de maní con oblea tipo golosina	Peanut nougat with candy wafer	A173	4
Tutuca	Puffed corn snack (tutuca)	A166	4
Vodka	Vodka	B125	4
Yogur con fitoesteroles	Yogurt with phytosterols	Y014	4

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Table S2. NOVA classification of ENNyS food and beverage codes (continued)

Item description (Spanish)	Item description (English)	Code	NOVA
Yogur descremado	Skim yogurt	Y003	4
Yogur descremado bebible	Drinkable skim yogurt	Y005	4
Yogur descremado con cereales	Skim yogurt with cereals	Y012	4
Yogur descremado con frutas	Skim yogurt with fruit	Y013	4
Yogur descremado con frutas y cereales	Skim yogurt with fruits and cereals	Y017	4
Yogur descremado fortificado con calcio	Calcium-fortified skim yogurt	Y015	4
Yogur entero bebible saborizado	Flavored Drinkable Whole Yogurt	Y004	4
Yogur entero con cereales	Whole yogurt with cereals	Y010	4
Yogur entero con frutas	Whole yogurt with fruits	Y011	4
Yogur entero con frutas y cereales	Whole yogurt with fruits and cereals	Y018	4
Yogur entero saborizado	Whole flavored yogurt	Y002	4
Yogur fortificado con hierro	Iron-fortified yogurt	Y016	4
Zapallo / calabaza, puré instantáneo (polvo)*	Pumpkin / squash, instant puree (powder)*	V084	4

Note. This table reports the 1,040 non-supplement ENNyS food and beverage codes classified by NOVA group. English item descriptions are provided as reference translations of the original Spanish labels used in the survey files.

Table S3. NOVA classification distribution (excluding supplements)

NOVA group	N items	% of items
NOVA 1	310	29.8
NOVA 2	69	6.6
NOVA 3	203	19.5
NOVA 4 (UPFs)	458	44.0
Total	**1,040**	**100.0**

Note. N items denotes the number of ENNyS food codes classified in each NOVA group. UPFs denotes ultra-processed foods.

Table S4. Indirect economic costs (PVLE) by sex and age group (intl.$ PPP)

Sex	Age	Deaths attr	PVLE per death	Total PVLE
Men	30–34	270	319k	86.0M
Men	35–39	254	291k	73.9M
Men	40–44	314	290k	91.0M
Men	45–49	429	229k	98.3M
Men	50–54	579	213k	123.6M
Men	55–59	776	108k	84.0M
Men	60–64	1,279	40k	50.6M
Men	65–69	1,798	0	0
Men subtotal	–	**5,699**	**107k**	**607.5M**
Women	30–34	137	182k	25.0M
Women	35–39	166	168k	27.8M
Women	40–44	241	170k	41.0M
Women	45–49	301	135k	40.6M
Women	50–54	435	111k	48.2M
Women	55–59	648	56k	36.0M
Women	60–64	926	12k	11.0M
Women	65–69	1,270	0	0
Women subtotal	–	**4,124**	**56k**	**229.6M**
Total	–	**9,823**	**85k**	**837.0M**

Note. Deaths attr denotes attributable deaths. PVLE denotes present value of lifetime earnings lost, PPP denotes purchasing power parity, and intl.$ denotes international dollars.

Table S5. Environmental mapping coverage

Scope	Rows covered (%)	Grams covered (%)	Energy covered (%)
All foods/beverages in-scope (excluding plain water/ice)	93.4	99.7	99.5
UPF only (NOVA 4)	96.5	99.8	99.6

Note. Coverage percentages indicate the share of rows, grams, and energy linked to environmental coefficients. UPF denotes ultra-processed food and NOVA 4 is the UPF group.

Table S6. Environmental replacement matrix for NDG policy scenarios (net % vs baseline)

Scenario	Structure	Basket	GHG	Land	Water	Eutro
Panel A. No substitution (reference policy scenarios)						
NDG conservative	No substitution	–	-6.7	-6.1	-13.6	-10.9
NDG central	No substitution	–	-7.8	-7.1	-15.8	-12.7
NDG strict	No substitution	–	-8.9	-8.1	-18.0	-14.5
Panel B. Isocaloric substitution (replacement matrix)						
NDG conservative	Isocaloric	NDG-concordant	0.0	-1.1	+21.0	+1.8
NDG conservative	Isocaloric	NOVA1	+7.0	+4.1	+16.1	+12.0
NDG conservative	Isocaloric	NOVA3	-5.0	-4.8	-9.0	-7.3
NDG conservative	Isocaloric	NOVA1/NOVA3 mix	-0.1	-1.1	+1.3	+0.6
NDG central	Isocaloric	NDG-concordant	0.0	-1.2	+24.4	+2.1
NDG central	Isocaloric	NOVA1	+8.1	+4.8	+18.7	+13.9
NDG central	Isocaloric	NOVA3	-5.8	-5.5	-10.4	-8.5
NDG central	Isocaloric	NOVA1/NOVA3 mix	-0.1	-1.3	+1.5	+0.7
NDG strict	Isocaloric	NDG-concordant	0.0	-1.4	+27.8	+2.4
NDG strict	Isocaloric	NOVA1	+9.3	+5.5	+21.4	+15.9
NDG strict	Isocaloric	NOVA3	-6.6	-6.3	-11.9	-9.7
NDG strict	Isocaloric	NOVA1/NOVA3 mix	-0.1	-1.5	+1.7	+0.7
Panel C. Isoweight substitution (replacement matrix)						
NDG conservative	Isoweight	NDG-concordant	+5.8	+5.9	+10.0	+10.4
NDG conservative	Isoweight	NOVA1	+5.5	+3.0	+12.8	+9.5
NDG conservative	Isoweight	NOVA3	-3.0	-3.2	-3.5	-3.0
NDG conservative	Isoweight	NOVA1/NOVA3 mix	+0.5	-0.7	+3.2	+2.1
NDG central	Isoweight	NDG-concordant	+6.7	+6.9	+11.7	+12.1
NDG central	Isoweight	NOVA1	+6.4	+3.5	+15.0	+11.1
NDG central	Isoweight	NOVA3	-3.5	-3.7	-4.1	-3.5
NDG central	Isoweight	NOVA1/NOVA3 mix	+0.6	-0.8	+3.7	+2.4
NDG strict	Isoweight	NDG-concordant	+7.7	+7.8	+13.3	+13.7
NDG strict	Isoweight	NOVA1	+7.3	+4.0	+17.1	+12.6
NDG strict	Isoweight	NOVA3	-3.9	-4.2	-4.7	-4.0
NDG strict	Isoweight	NOVA1/NOVA3 mix	+0.7	-0.9	+4.2	+2.8

Note. Values are net percentage changes versus baseline. GHG denotes greenhouse gas emissions, Eutro denotes freshwater eutrophication, and NDG denotes National Dietary Guidelines. Negative values indicate reductions and positive values indicate increases.

Table S7. Diet composition of the scenario baskets and UPF targets used in the analyses

Panel A. Diet composition of replacement baskets (energy share, %)

Food group	Observed non-UPF	NOVA1	NOVA3	NOVA1/NOVA3 mix	NDG target
Fruits	7.3	13.7	5.7	9.0	18.0
Vegetables (outdoor)	3.3	7.8	1.4	4.0	25.0
Legumes and pulses	0.4	1.2	0.1	0.5	5.0
Cereals (without rice)	8.0	7.1	2.0	4.0	6.0
Rice	4.5	13.0	0.0	5.3	3.0
Pasta	6.2	0.0	13.2	7.8	3.0
Baked products	25.8	0.0	55.4	32.8	3.0
Starchy vegetables	0.2	0.7	0.0	0.3	3.0
Dairy products	10.0	5.0	15.4	11.1	13.0
Beef	3.3	7.1	0.6	3.3	4.0
Lamb and mutton	0.3	1.0	0.0	0.4	0.5
Other meats	0.0	0.0	0.0	0.0	0.5
Poultry	7.4	21.6	0.0	8.8	2.5
Pork	6.1	10.5	5.2	7.4	1.0
Fish and seafood	0.9	1.8	0.5	1.0	3.0
Eggs	2.9	8.3	0.0	3.4	1.5
Oil crops	13.4	1.2	0.4	0.7	6.0
Nuts and seeds	0.1	0.0	0.2	0.1	2.0

Panel B. Scenario-level UPF targets

Scenario	Target UPF (% energy)	Reduction vs baseline (%)
Observed baseline	16.66	0.0
10% proportional reduction	14.99	10.0
20% proportional reduction	13.33	20.0
50% proportional reduction	8.33	50.0
NDG conservative target	5.45	67.3
NDG central target	3.63	78.2
NDG strict target	1.81	89.1

Note. **Panel A.** Each column reports the share of total energy contributed by each food group *within* that basket; columns sum to 100%. **Observed non-UPF** is the actual energy composition of all non-UPF foods consumed (NOVA 1, 2, and 3 combined) by adults aged 30–69 in ENNyS 2, restricted to foods with environmental coefficients and excluding plain drinking water. This column is an empirical reference and is *not* used as a replacement basket in the counterfactual scenarios. **NOVA 1** and **NOVA 3** show the energy composition within minimally processed and processed non-UPF foods, respectively, based on observed ENNyS 2 intakes. In the current Argentine data, the NOVA 1 basket is concentrated in animal-source foods together with fruits and rice, whereas the NOVA 3 basket is dominated by baked products, dairy, pasta, and other cereal-based items. **NOVA 1/NOVA 3 mix** is a weighted average of those two baskets, with weights derived from their respective model-mapped energy totals (approximately 39% NOVA 1 and 61% NOVA 3). **NDG target** shows the normative food-group proportions prescribed by Argentina's National Dietary Guidelines (NDG 2016)—these are guideline recommendations, not observed intakes, and differ substantially from the current diet (notably higher in fruits and vegetables, lower in baked products and oil crops). **Panel B.** "Target UPF (% energy)" is the counterfactual share of ultra-processed foods in total dietary energy; "Reduction vs baseline (%)" is the proportional reduction from the observed baseline reported in Panel B. Proportional scenarios (10%, 20%, 50%) scale the UPF energy share downward uniformly without altering the food-group composition of the non-UPF diet. NDG scenarios impose the NDG basket composition from Panel A and derive the residual UPF allowance from the guideline food-group targets, yielding substantially larger reductions (67–89%) because Argentina's current UPF share is far above the level implied by guideline-adherent dietary patterns. The three NDG profiles (conservative, central, strict) correspond to progressively tighter guideline adherence, yielding target UPF shares of 5.4%, 3.6%, and 1.8%, respectively.

Table S8. Environmental coefficients by food group

Food group	GHG (kg CO_2-eq/kg)	Land (m^2/kg)	Water (L/kg)	Eutro (g PO_4-eq/kg)	Boundary	FW/RW (%)	Source
Baked products	0.71	5.45	11.00	4.62	Farm gate	119.27	Arrieta et al. 2022 (Table S4, Part I)
Beef	40.93	251.63	130.77	42.53	Farm gate	72.00	Arrieta et al. 2022 (Table S4, Part I)
Beer	1.27	0.87	82.20	2.78	Retail/product	100.00	Poore and Nemecek 2018 (retail proxy)
Cereals (without rice)	0.82	4.60	435.00	3.90	Farm gate	65.00	Arrieta et al. 2022 (Table S4, Part I)
Coffee	28.53	21.62	25.90	110.52	Retail/product	100.00	OWID / Poore and Nemecek 2018
Confectionery/candy	2.10	2.50	180.00	7.50	Retail/product	100.00	Agribalyse 3.1 proxy
Dairy products	1.14	2.98	25.00	11.98	Farm gate	105.00	Arrieta et al. 2022 (Table S4, Part I)
Eggs	1.32	6.20	57.00	12.07	Farm gate	100.00	Arrieta et al. 2022 (Table S4, Part I)
Fish and seafood	1.83	0.01	0.00	2.45	Farm gate	51.00	Arrieta et al. 2022 (Table S4, Part I)
Flavoured water drink	0.30	0.12	12.00	0.55	Retail/product	100.00	Agribalyse 3.1 proxy
Fruits	0.18	0.40	239.00	10.00	Farm gate	51.00	Arrieta et al. 2022 (Table S4, Part I)
Ice cream	2.45	5.80	280.00	14.20	Retail/product	100.00	Agribalyse 3.1 proxy
Lamb and mutton	35.80	127.80	189.57	352.87	Farm gate	71.00	Arrieta et al. 2022 (Table S4, Part I)
Legumes and pulses	0.22	4.30	33.00	2.50	Farm gate	78.00	Arrieta et al. 2022 (Table S4, Part I)
Mate/yerba	1.20	2.10	45.00	3.50	Retail/product	100.00	Argentina-specific estimate
Nuts and seeds	2.01	0.33	852.00	3.62	Farm gate	50.00	Arrieta et al. 2022 (Table S4, Part I)
Oil crops	1.72	10.70	11.00	18.36	Farm gate	29.00	Arrieta et al. 2022 (Table S4, Part I)
Other meats	35.80	127.80	189.57	352.87	Farm gate	71.00	Arrieta et al. 2022 (Table S4, Part I)
Pasta	0.71	5.45	11.00	4.62	Farm gate	59.21	Arrieta et al. 2022 (Table S4, Part I)
Pork	2.24	9.86	238.67	16.33	Farm gate	68.00	Arrieta et al. 2022 (Table S4, Part I)
Poultry	0.99	6.87	75.00	16.34	Farm gate	75.00	Arrieta et al. 2022 (Table S4, Part I)
Rice	1.68	1.80	448.00	10.00	Farm gate	49.00	Arrieta et al. 2022 (Table S4, Part I)
Soft drinks (carbonated)	0.41	0.16	17.50	0.82	Retail/product	100.00	Agribalyse 3.1 proxy
Spirits (distilled)	2.80	1.20	82.00	5.10	Retail/product	100.00	Agribalyse 3.1 proxy
Starchy vegetables	0.13	0.33	51.00	1.20	Farm gate	63.00	Arrieta et al. 2022 (Table S4, Part I)
Sugars	2.50	1.94	418.90	11.16	Retail/product	100.00	OWID / Poore and Nemecek 2018
Tea/infusions	0.52	0.28	15.20	1.85	Retail/product	100.00	Poore and Nemecek 2018 proxy
Vegetables (outdoor)	0.34	0.48	140.00	0.61	Farm gate	51.00	Arrieta et al. 2022 (Table S4, Part I)
Wine	1.79	1.78	78.90	4.57	Retail/product	100.00	OWID / Poore and Nemecek 2018

Note. This table reports the coefficient set used in the manuscript environmental specification (curated run; main manuscript Table 5 and Online Appendix Tables S5–S6), restricted to food groups with mapped intake in adults aged 30–69 years. GHG denotes greenhouse gas emissions and Eutro denotes freshwater eutrophication. FW/RW is the farm-weight to retail-weight

factor used to transform observed retail intake to coefficient reference mass, with conversion $kg_{ref} = kg_{retail}/(FW/RW/100)$. FW/RW values of 100 imply no mass-conversion adjustment. Source labels refer to the exact provenance of each group-level value: Arrieta et al. 2022 (Argentina-focused LCA), OWID/Poore and Nemecek 2018, and documented proxy or estimated extensions.

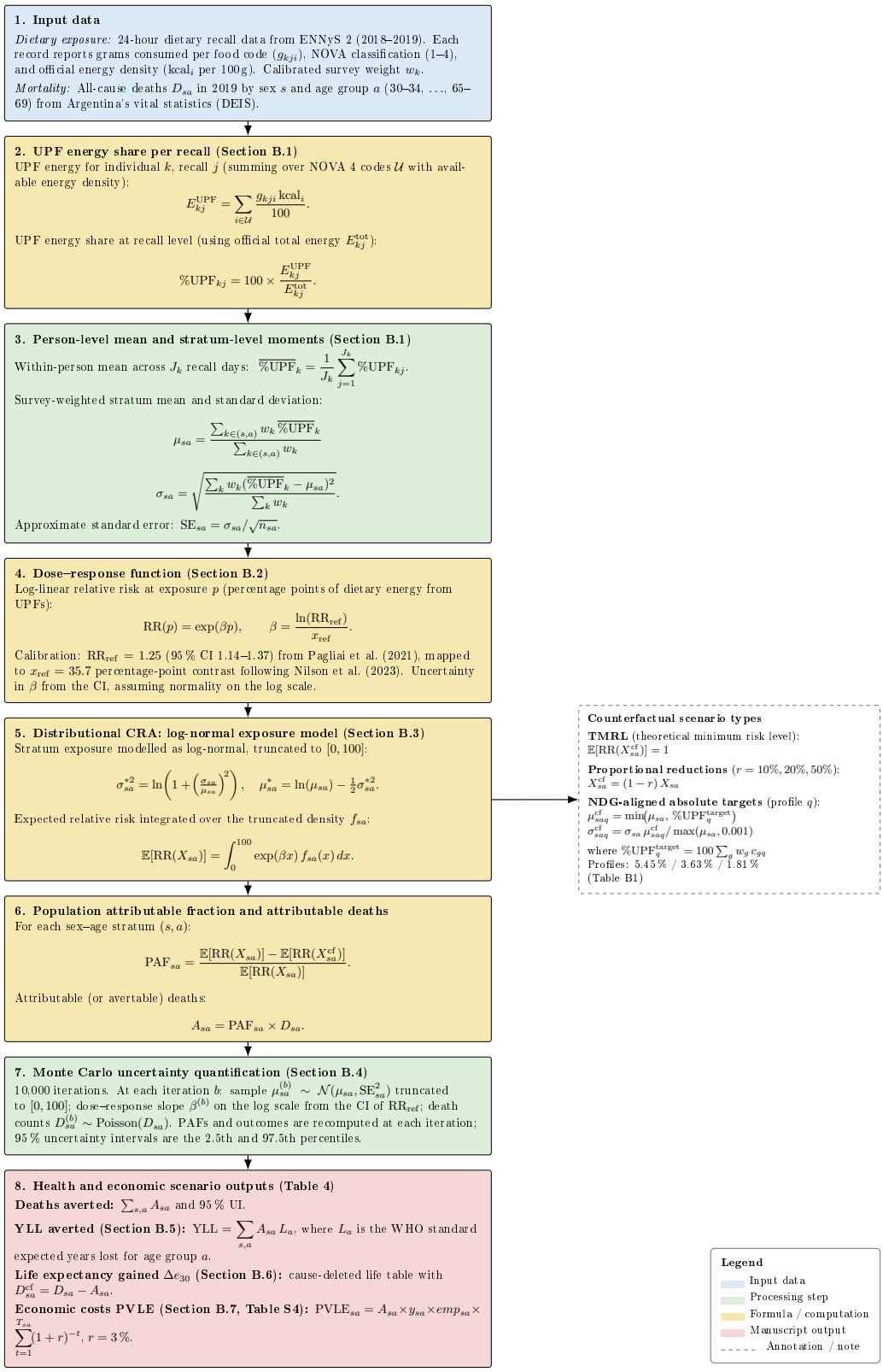

Fig S1. Flow diagram of health impact estimation. Blue = input data; green = processing; yellow = formula/computation; red = manuscript output. Note: solid arrows indicate the main computational sequence, while dashed arrows indicate auxiliary annotations. Section and table references point to this appendix.

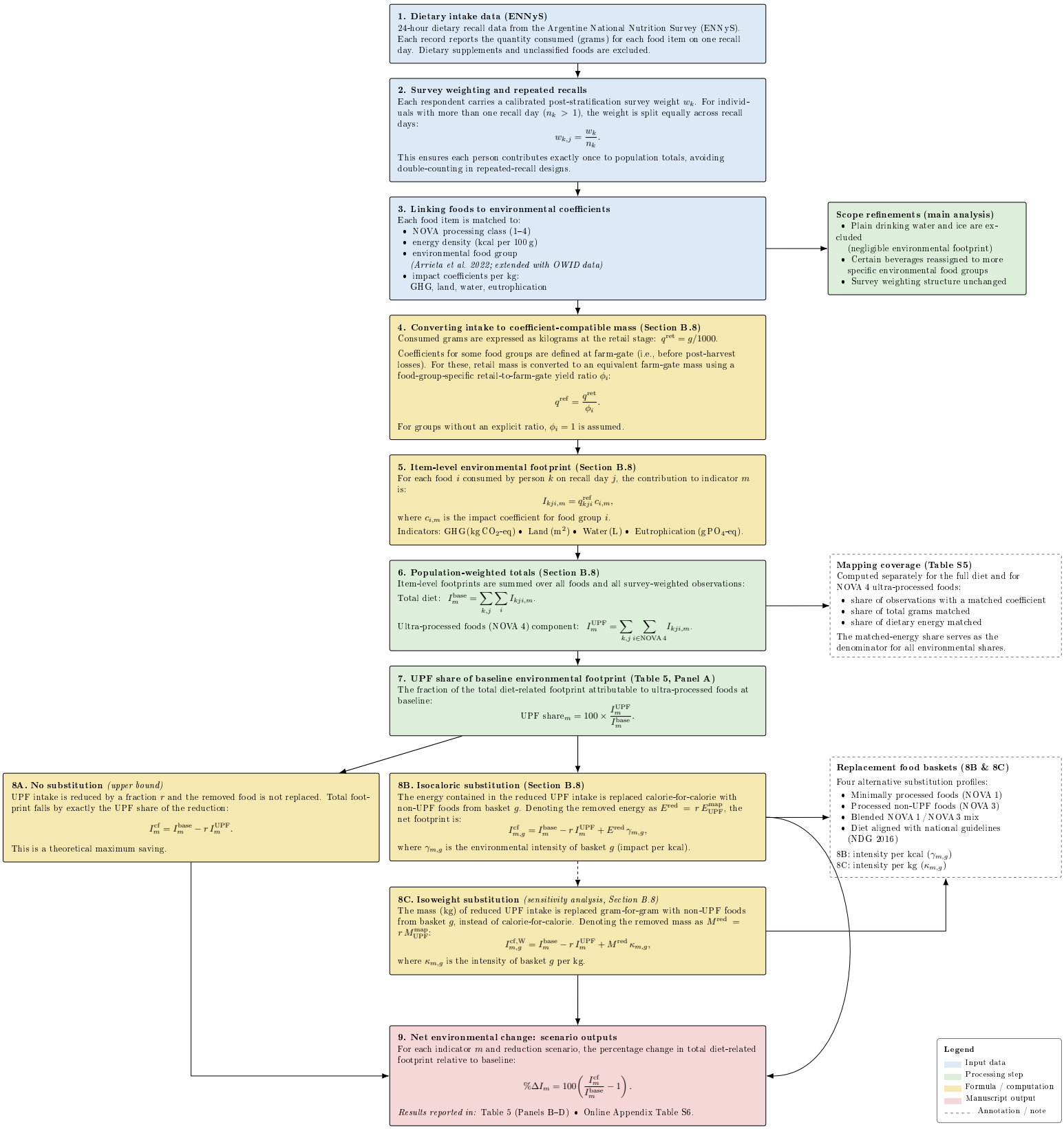

Fig S2. Flow diagram of environmental impact estimation. Blue = input data; green = processing; yellow = formula/computation; red = manuscript output. Note: the dashed arrow from 8B to 8C indicates that isoweight substitution is a sensitivity variant, not a sequential step.

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